# Selection Plots: Config Comparisons

April 8, 2026

## Contents


**1 bdt_nom vs bdt_tightbdt50** **2**

1.1 Sideband (Data) . . . 2
1.2 Sideband (MC) . . . 5
1.3 Purity . . . 7
1.4 Efficiency . . . 8
1.5 Final result . . . 11
1.6 Isolation fit (CONF) . . . 14
1.7 Isolation ET . . . 18

**2 bdt_nom vs bdt_npb03** **24**

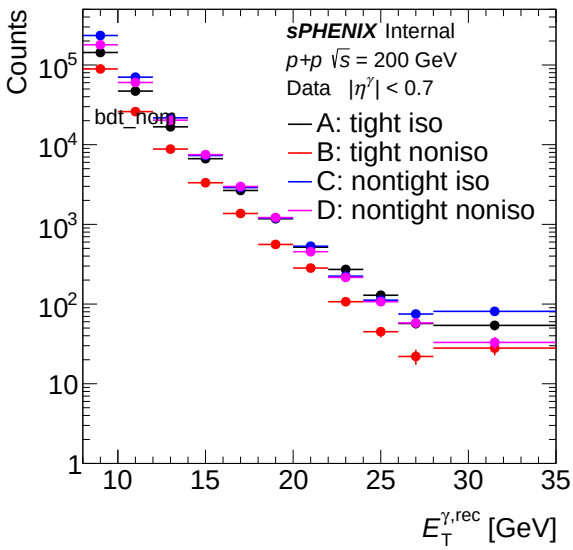
2.1 Sideband (Data) . . . 24
2.2 Sideband (MC) . . . 26
2.3 Purity . . . 28
2.4 Efficiency . . . 29
2.5 Final result . . . 32
2.6 Isolation fit (CONF) . . . 35
2.7 Isolation ET . . . 39

# 1 bdt_nom vs bdt_tightbdt50

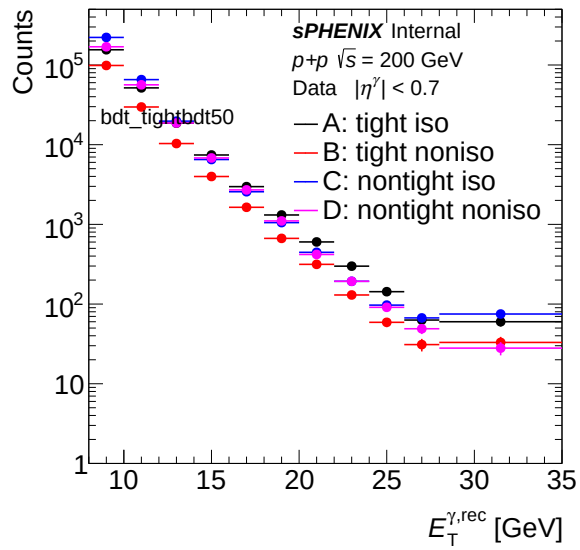
## Config diff (A → B)

```diff
--- config_bdt_nom.yaml
+++ config_bdt_tightbdt50.yaml
@@ -15,5 +15,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_tightbdt50"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -42,6 +42,6 @@
   eff_dR: 0.1
   trigger_used: 30
-  pT_bins: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30]
-  pT_bins_truth: [7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 35]
+  pT_bins: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 35]
+  pT_bins_truth: [7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 35, 40]
   iso_threshold: 1
   iso_hcalonly: 0
@@ -129,5 +129,5 @@
     #bdt
     bdt_max: 1.0
-    bdt_min: 0.6
+    bdt_min: 0.5


@@ -183,5 +183,5 @@

     #bdt
-    bdt_max: 0.6
+    bdt_max: 0.5
     bdt_min: 0.02
```

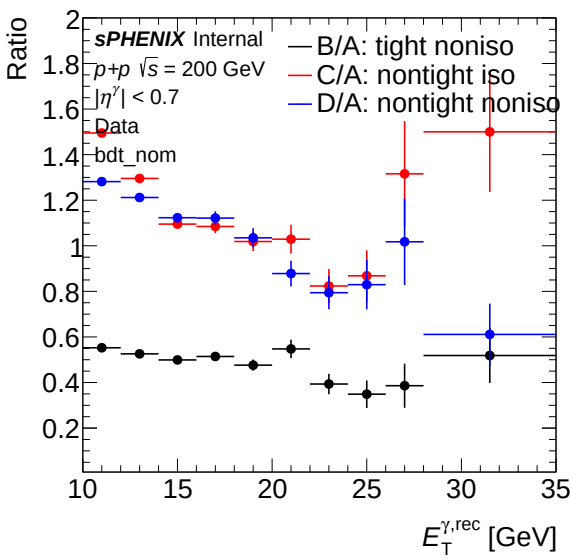
### 1.1 Sideband (Data)



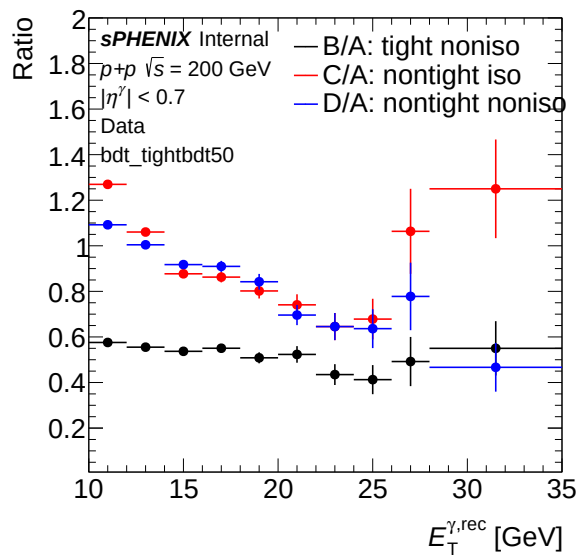

(a) Sideband ET spectrum (data) — bdt_nom



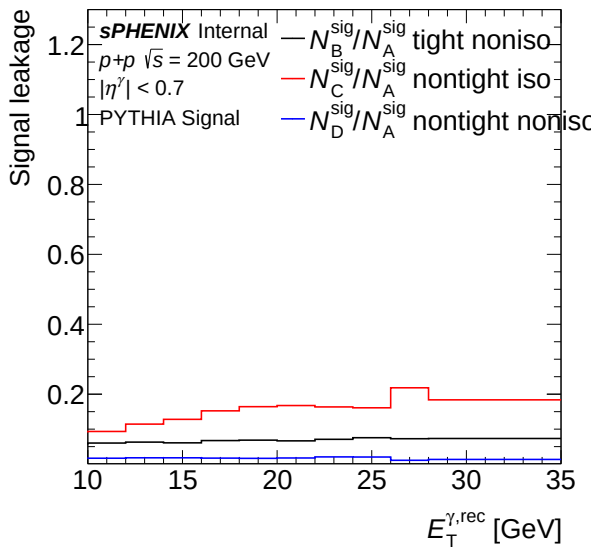

(b) Sideband ET spectrum (data) — bdt_tightbdt50



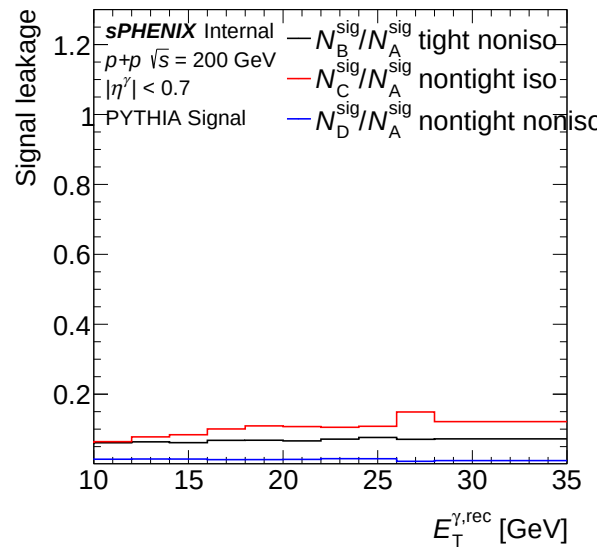

(a) Sideband ET ratio (data) — bdt_nom




(b) Sideband ET ratio (data) — bdt_tightbdt50

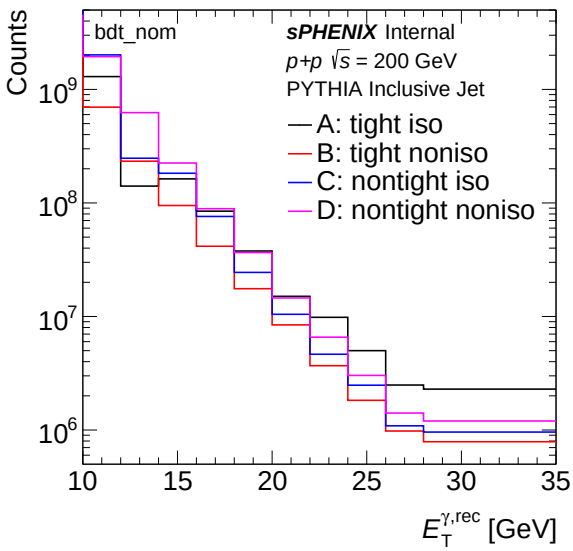



(a) Leakage fraction vs ET (data) — bdt_nom

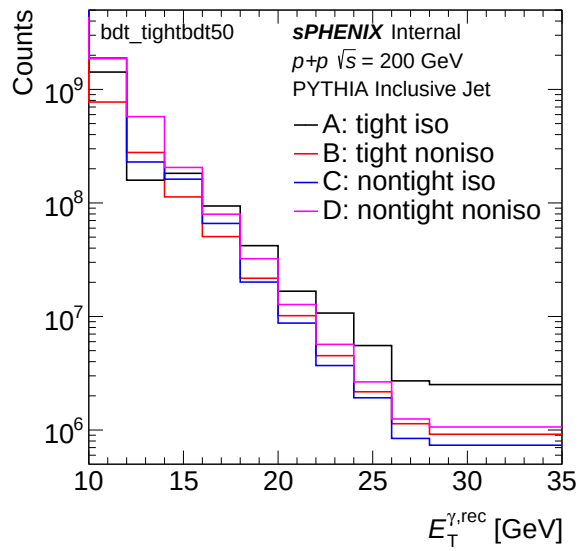



(b) Leakage fraction vs ET (data) — bdt_tightbdt50

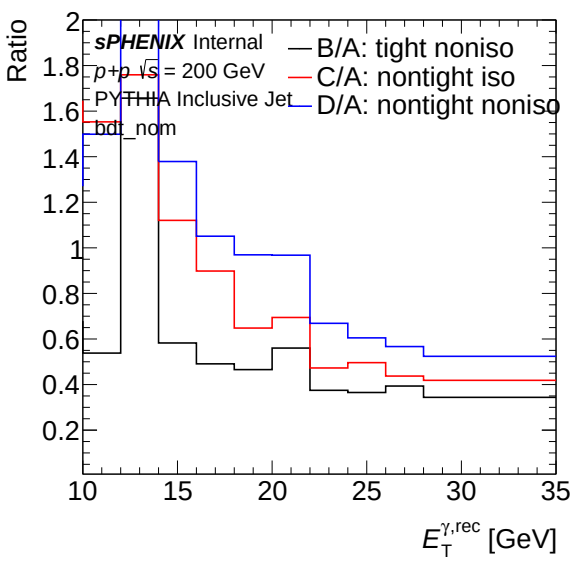
## 1.2 Sideband (MC)



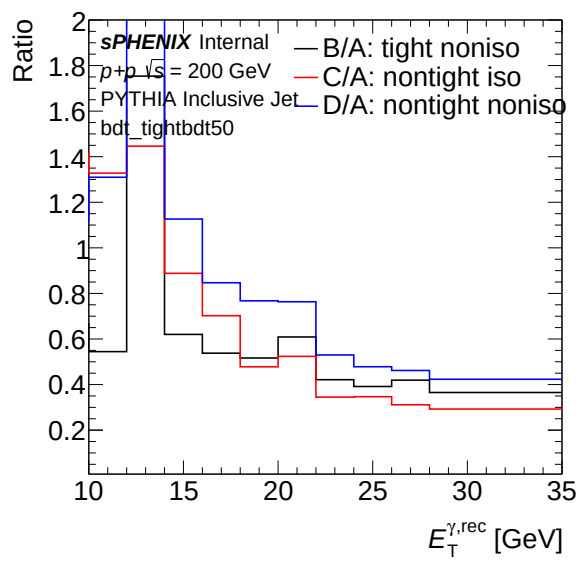

(a) Sideband ET spectrum (MC) — bdt_nom



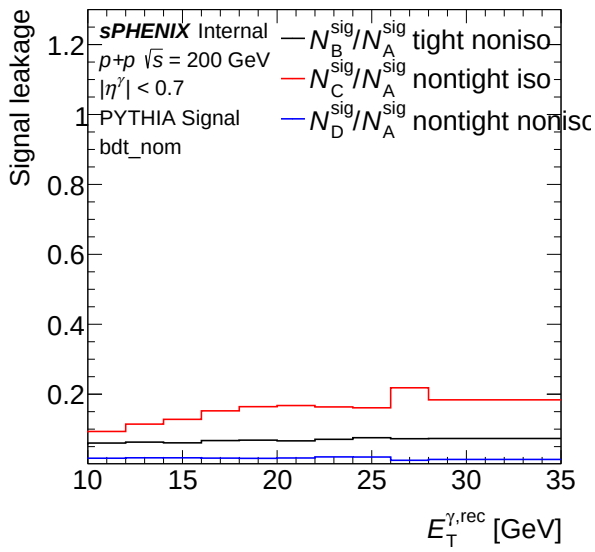

(b) Sideband ET spectrum (MC) — bdt_tightbdt50



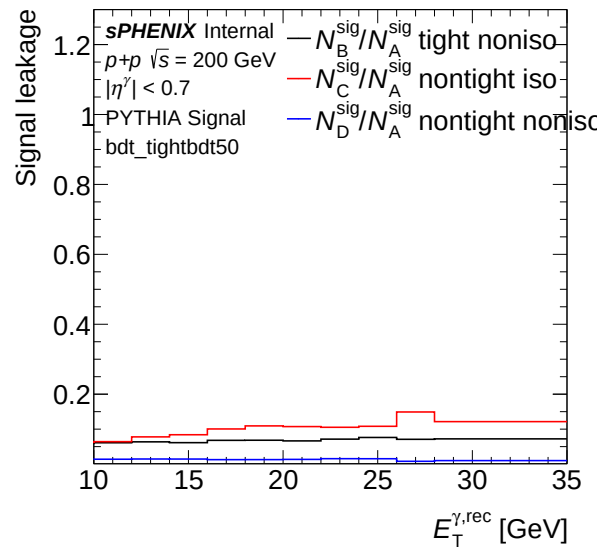

(a) Sideband ET ratio (MC) — bdt_nom




(b) Sideband ET ratio (MC) — bdt_tightbdt50

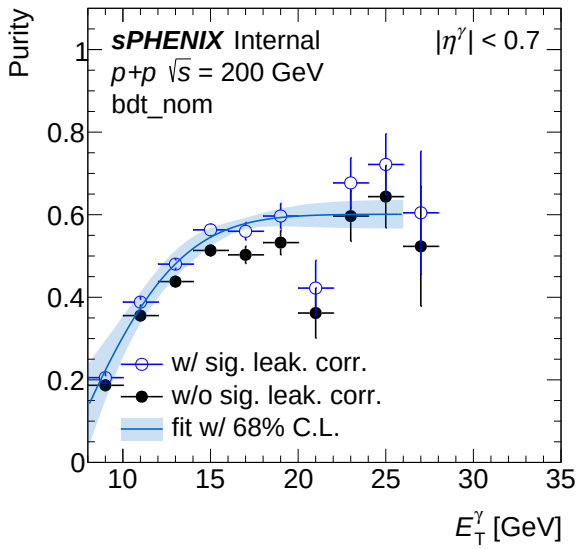



(a) Leakage fraction vs ET (MC) — bdt_nom

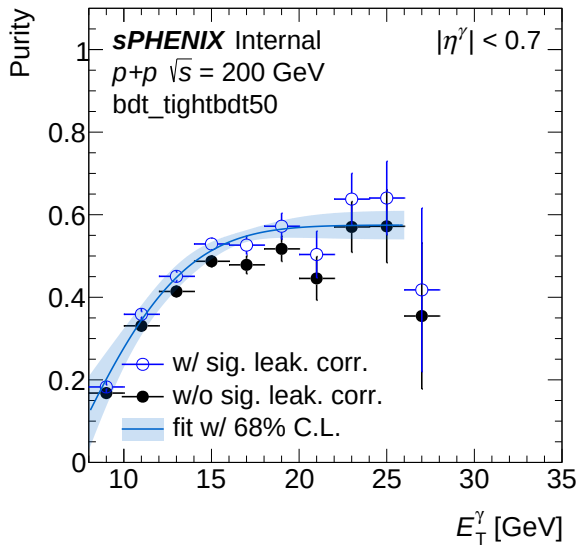



(b) Leakage fraction vs ET (MC) — bdt_tightbdt50

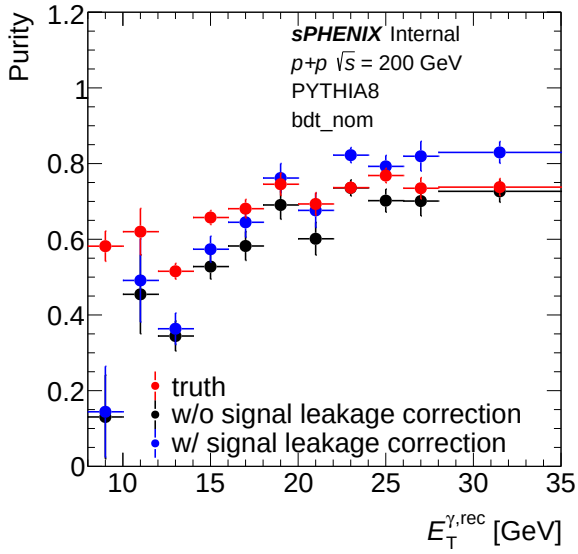
## 1.3 Purity



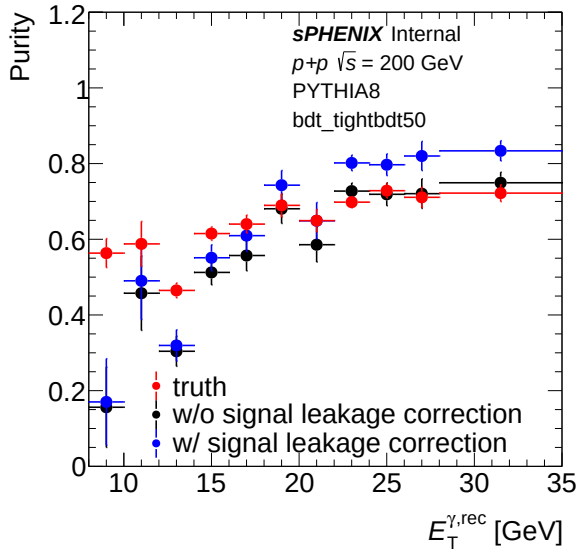

(a) Purity (data) — bdt_nom




(b) Purity (data) — bdt_tightbdt50

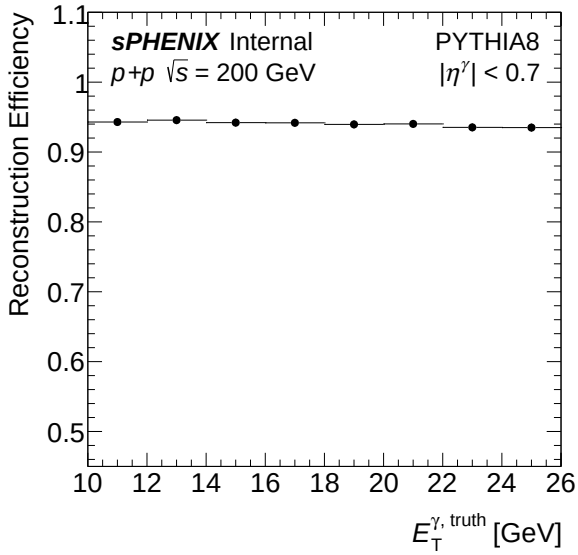



(a) Purity (MC) — bdt_nom

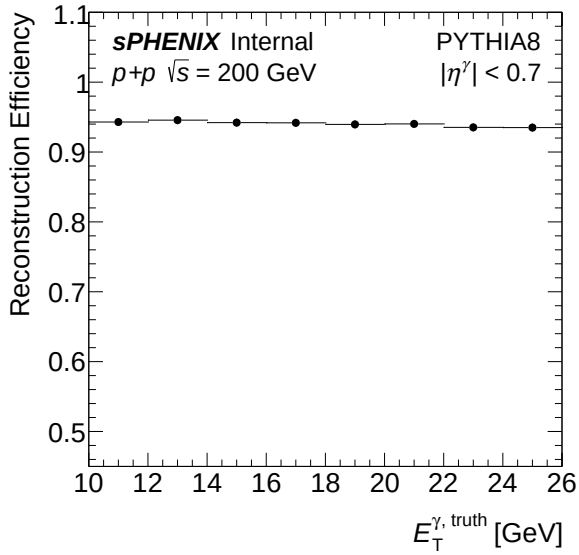



(b) Purity (MC) — bdt_tightbdt50

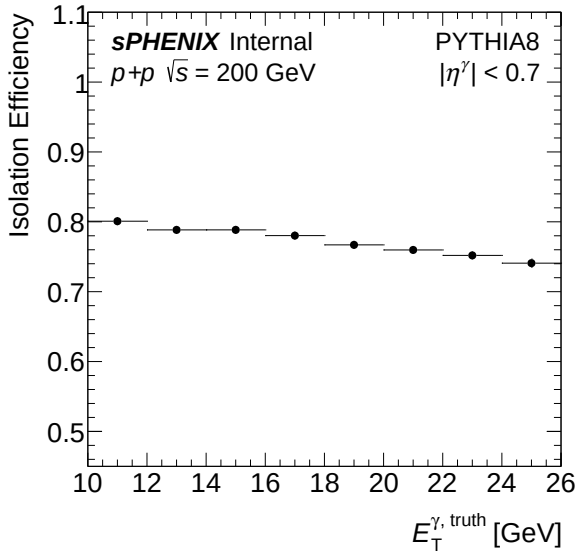
## 1.4 Efficiency



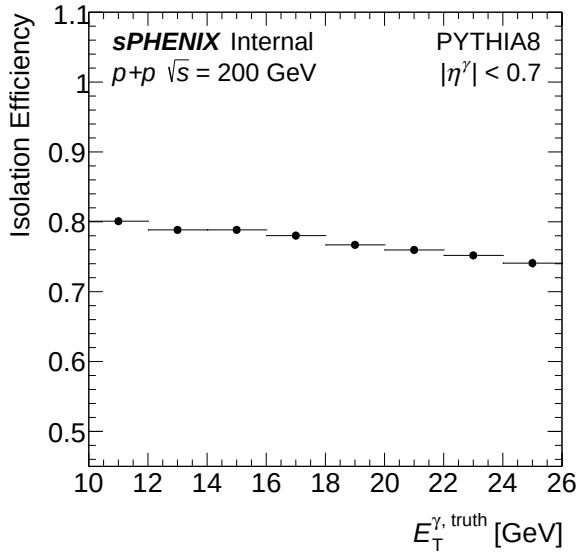

(a) Reconstruction efficiency — bdt_nom



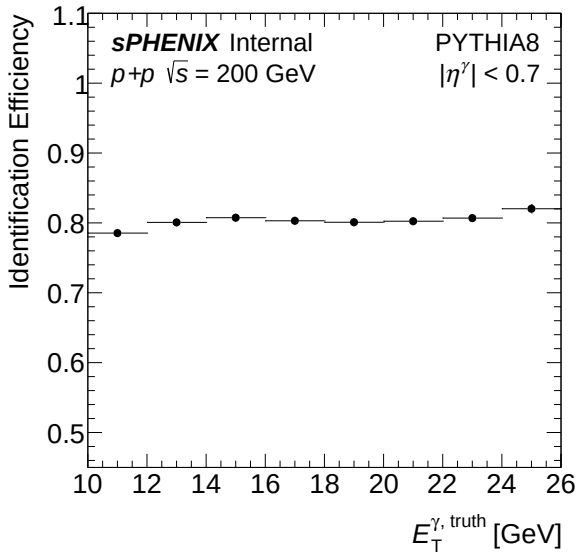

(b) Reconstruction efficiency — bdt_tightbdt50



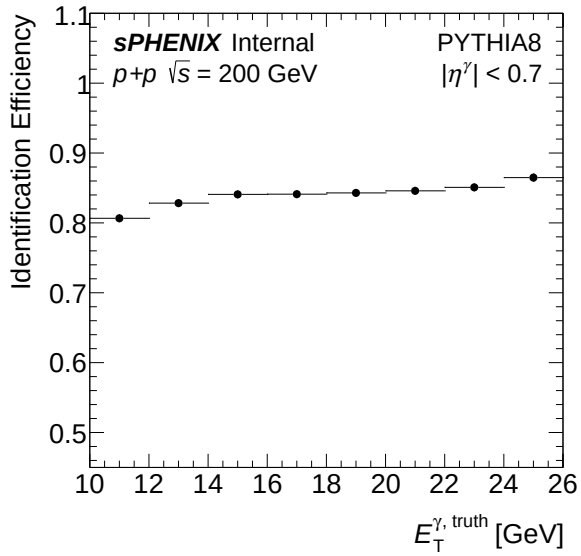

(a) Isolation efficiency — bdt_nom



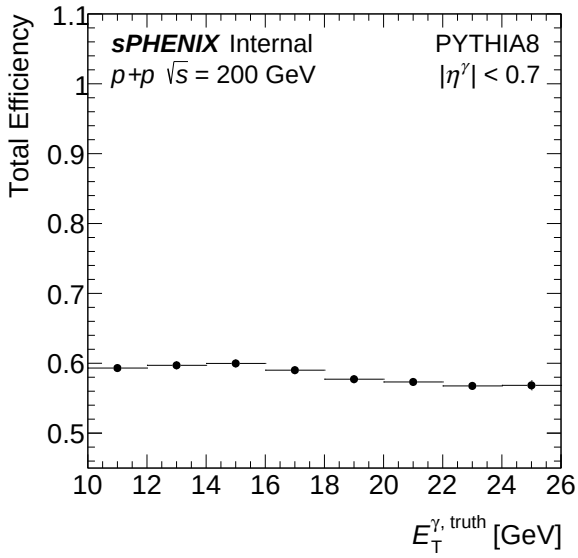

(b) Isolation efficiency — bdt_tightbdt50



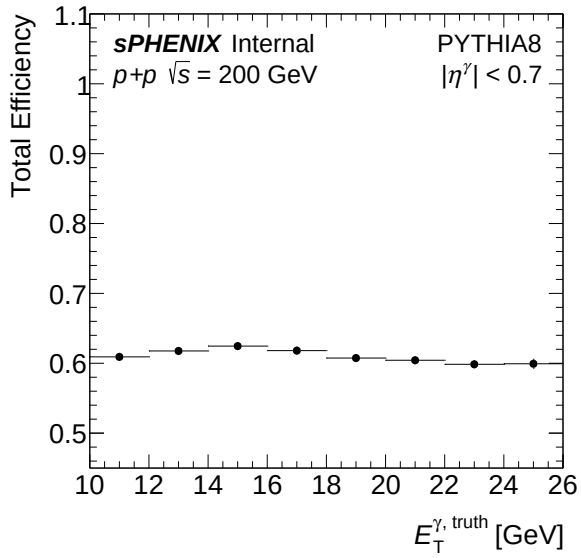

(a) Identification efficiency — bdt_nom



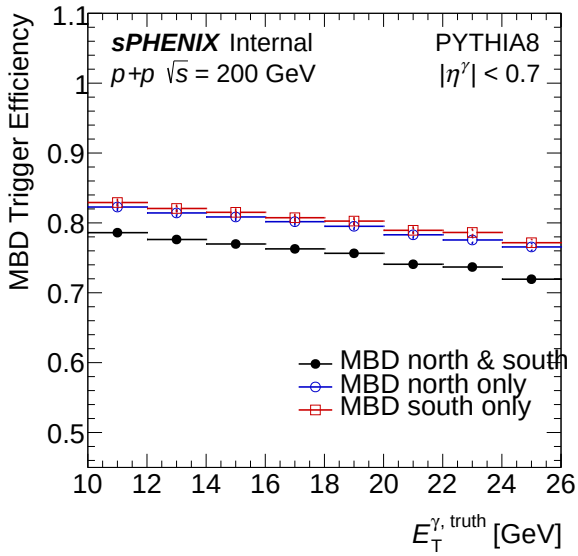

(b) Identification efficiency — bdt_tightbdt50



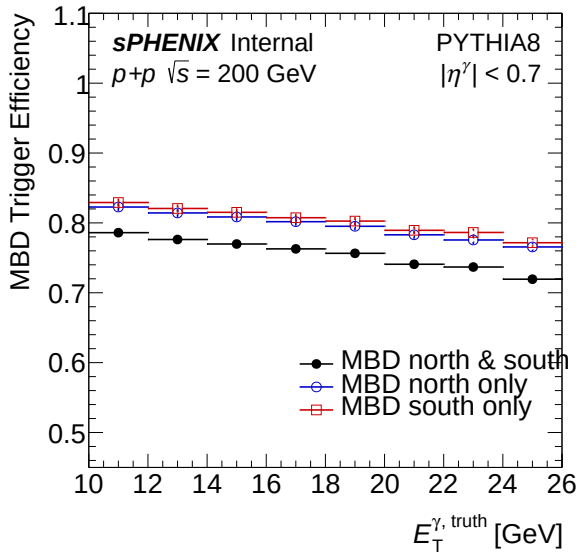

(a) Total efficiency — bdt_nom



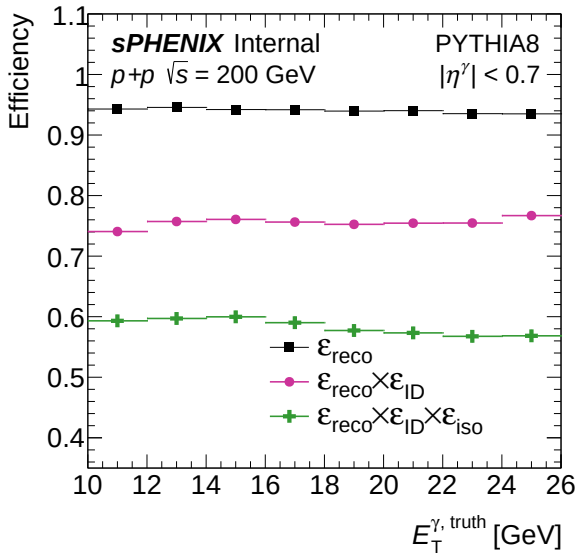

(b) Total efficiency — bdt_tightbdt50



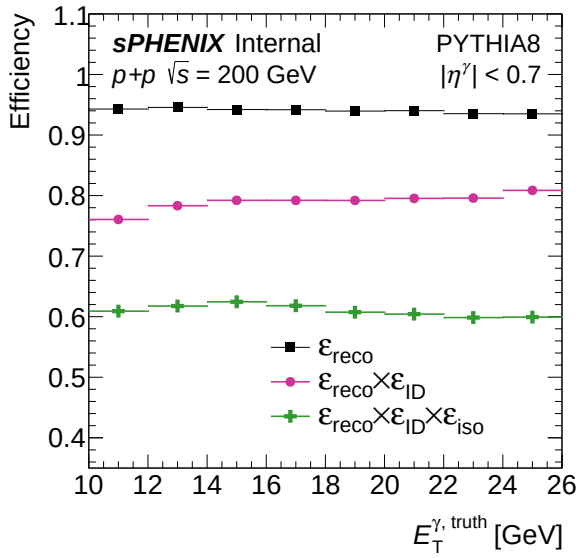

(a) MBD trigger efficiency — bdt_nom




(b) MBD trigger efficiency — bdt_tightbdt50

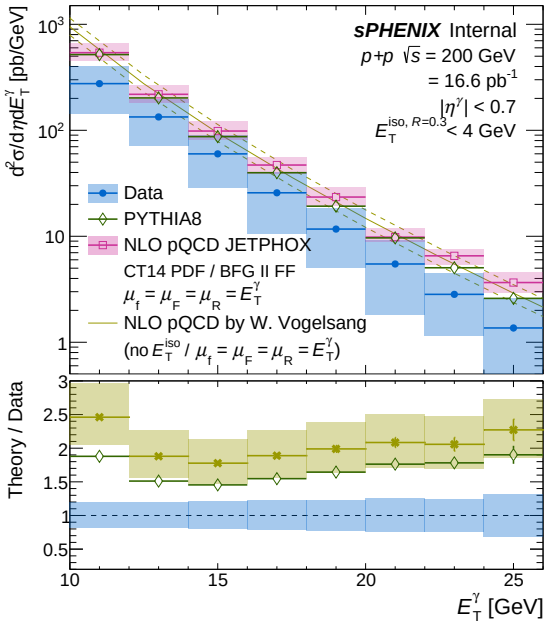



(a) Combined photon efficiency — bdt_nom

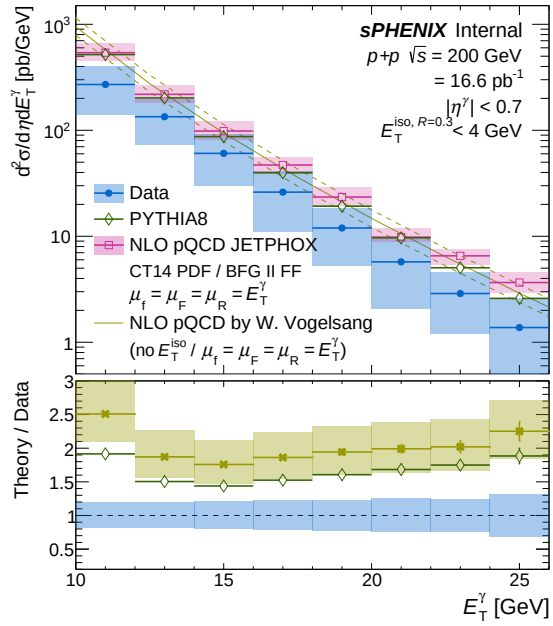



(b) Combined photon efficiency — bdt_tightbdt50

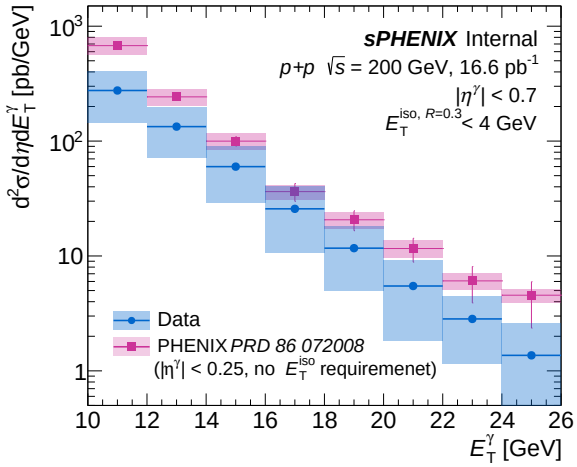
## 1.5 Final result



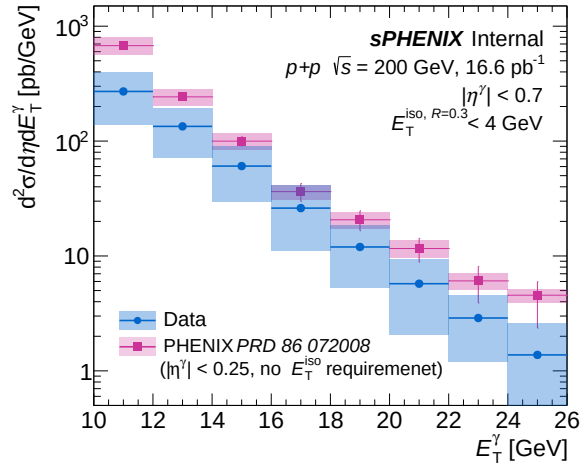

(a) Final cross section result — bdt_nom



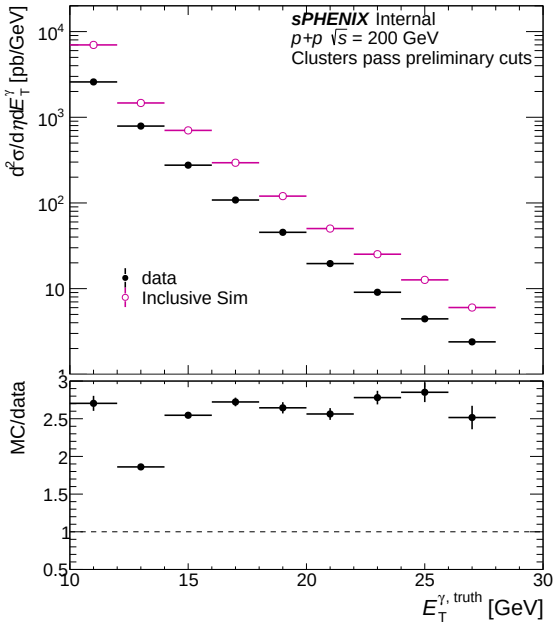

(b) Final cross section result — bdt_tightbdt50



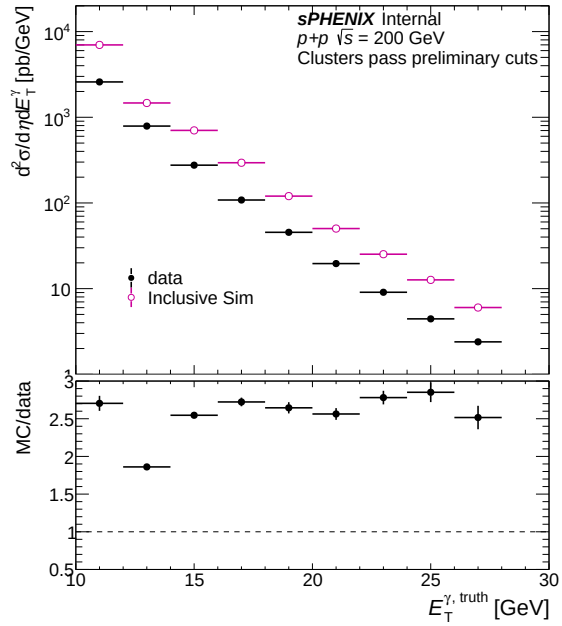

(a) Comparison with PHENIX — bdt_nom



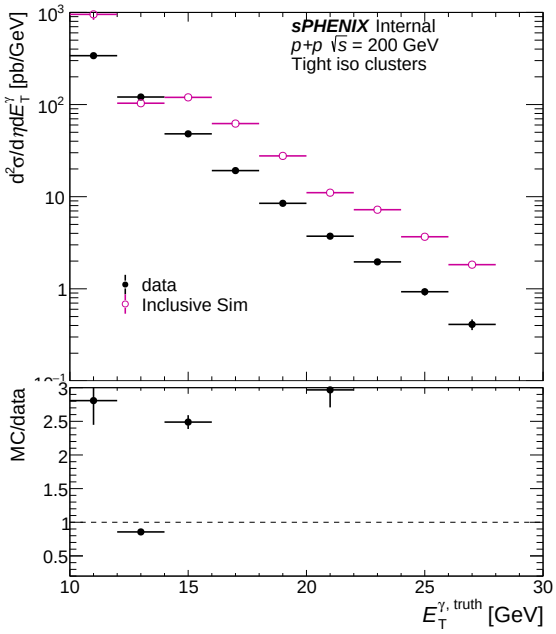

(b) Comparison with PHENIX — bdt_tightbdt50



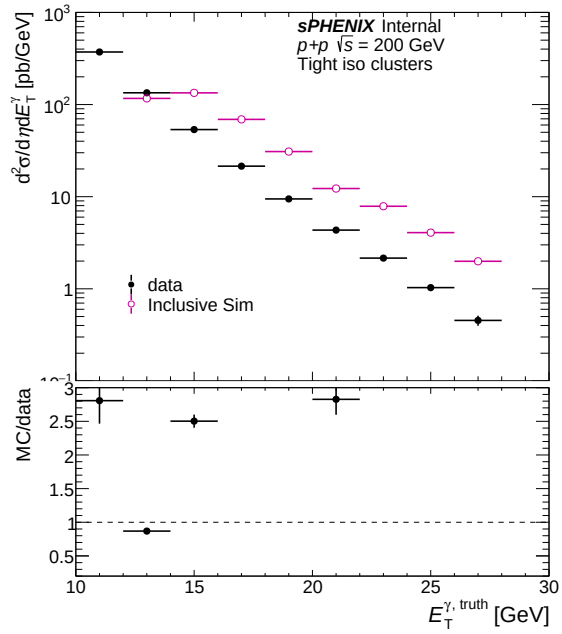

(a) Common cluster yield (data vs MC) — bdt_nom



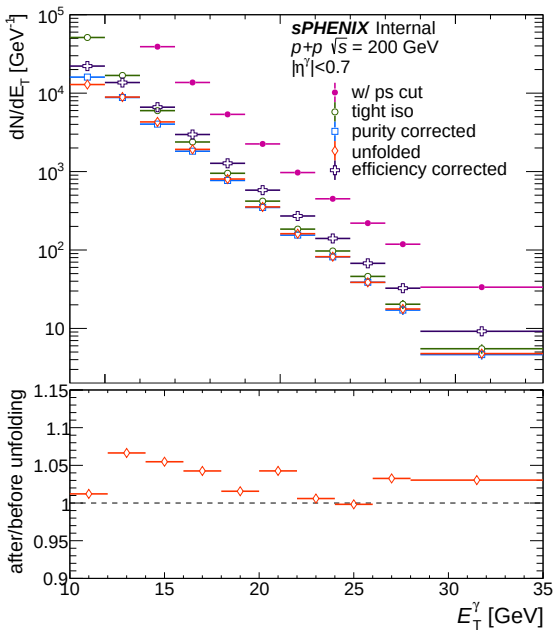

(b) Common cluster yield (data vs MC) — bdt_tightbdt50



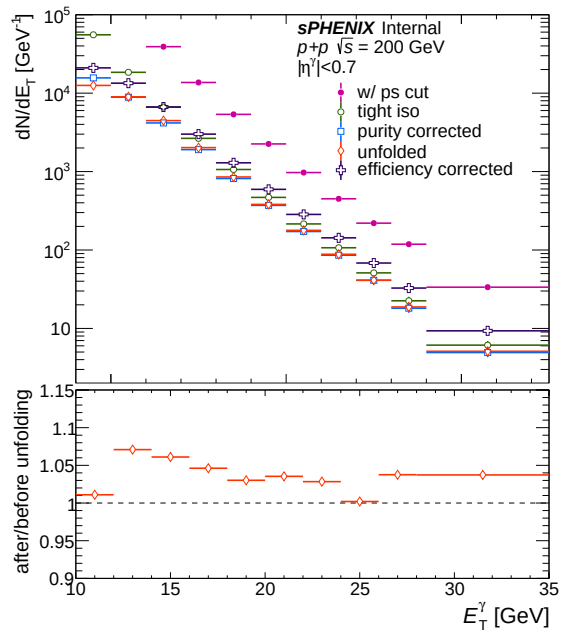

(a) Tight iso cluster yield (data vs MC) — bdt_nom




(b) Tight iso cluster yield (data vs MC) — bdt_tightbdt50

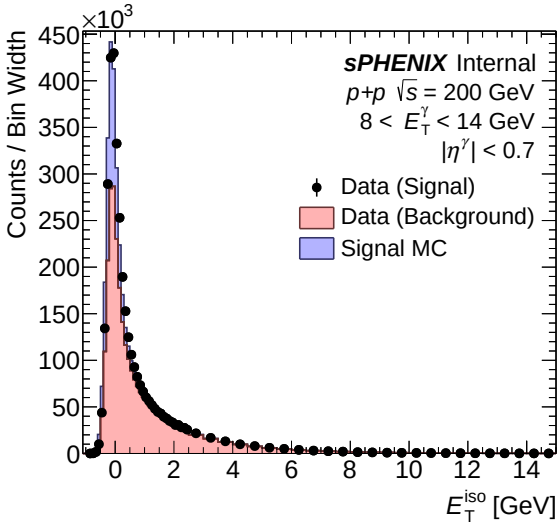



(a) All correction steps — bdt_nom

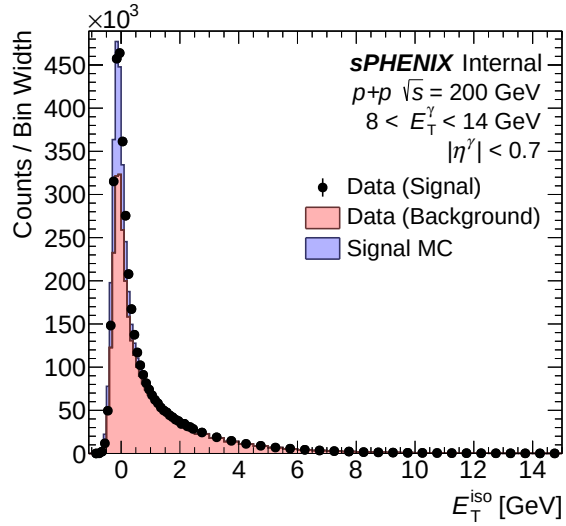



(b) All correction steps — bdt_tightbdt50

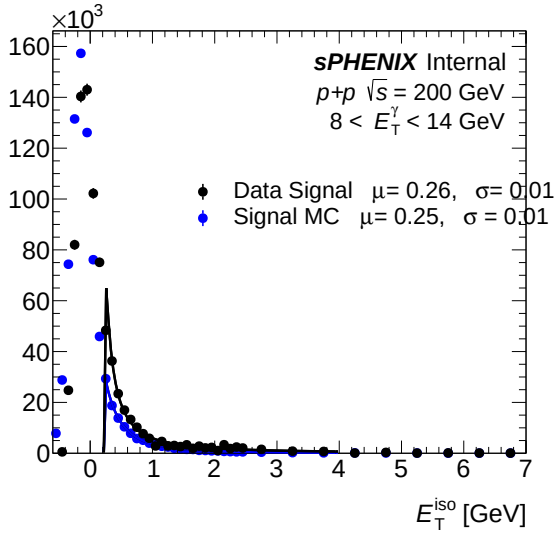
## 1.6 Isolation fit (CONF)



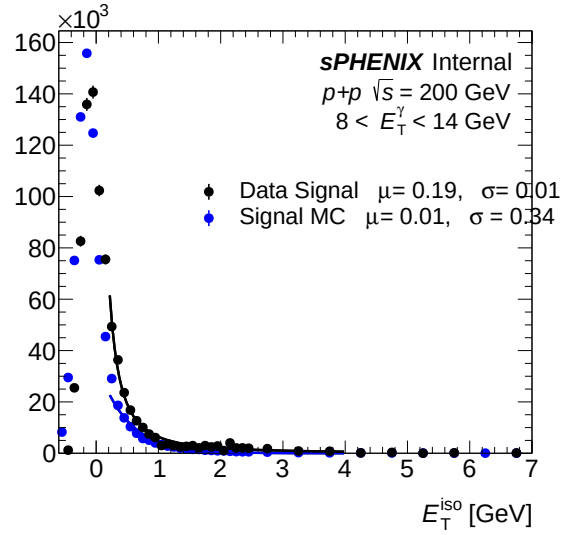

(a) IsoET template (pT bins 0–2) — bdt_nom



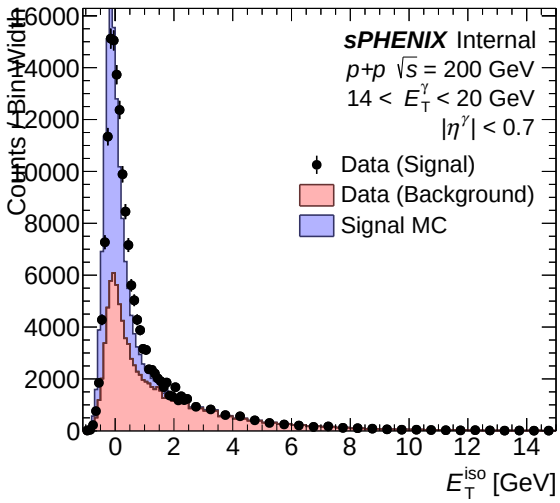

(b) IsoET template (pT bins 0–2) — bdt_tightbdt50



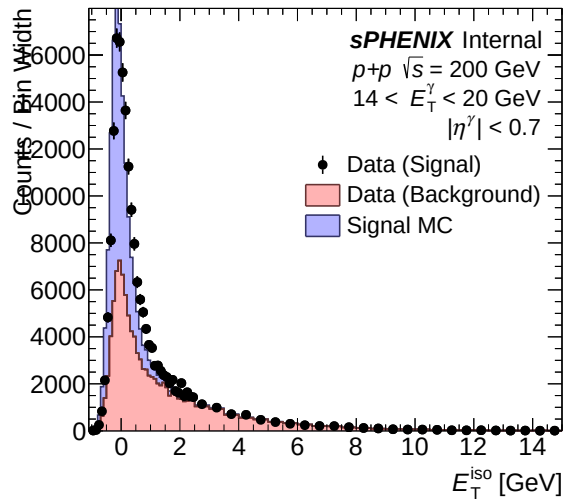

(a) IsoET fit template (pT bins 0–2) — bdt_nom



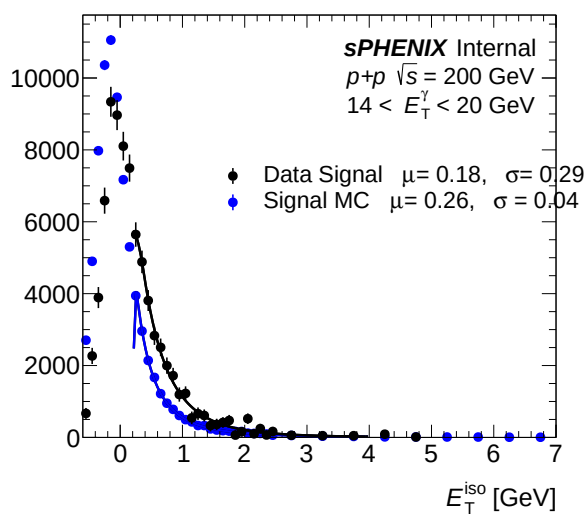

(b) IsoET fit template (pT bins 0–2) — bdt_tightbdt50



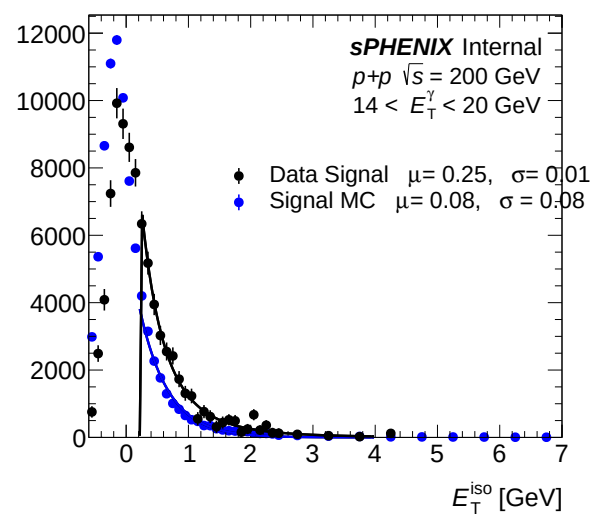

(a) IsoET template (pT bins 3–5) — bdt_nom



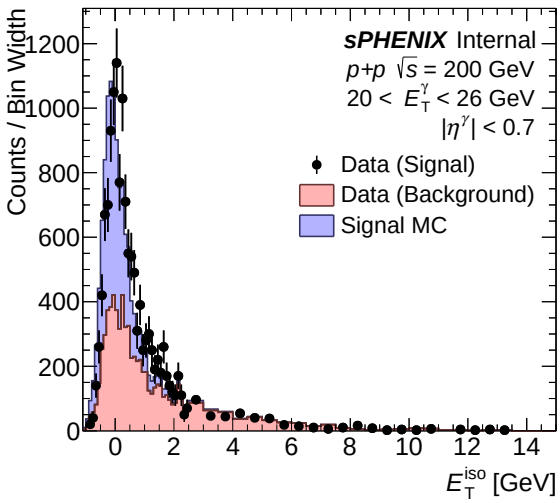

(b) IsoET template (pT bins 3–5) — bdt_tightbdt50



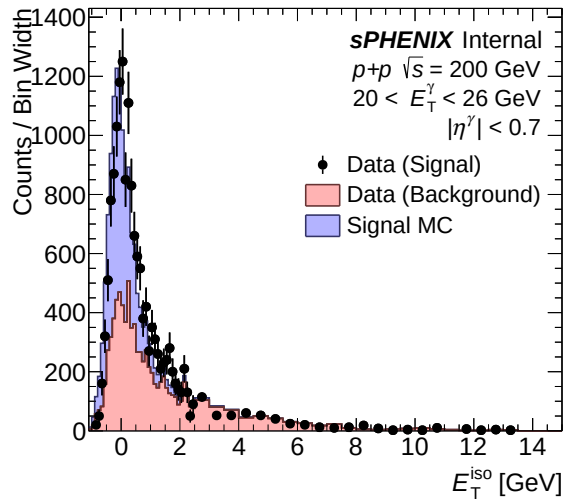

(a) IsoET fit template (pT bins 3–5) — bdt_nom



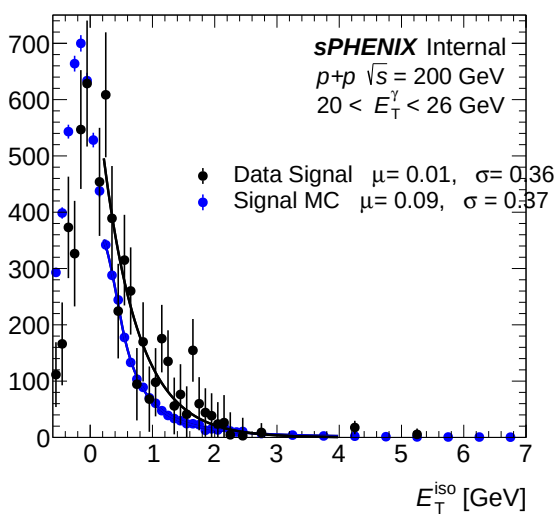

(b) IsoET fit template (pT bins 3–5) — bdt_tightbdt50



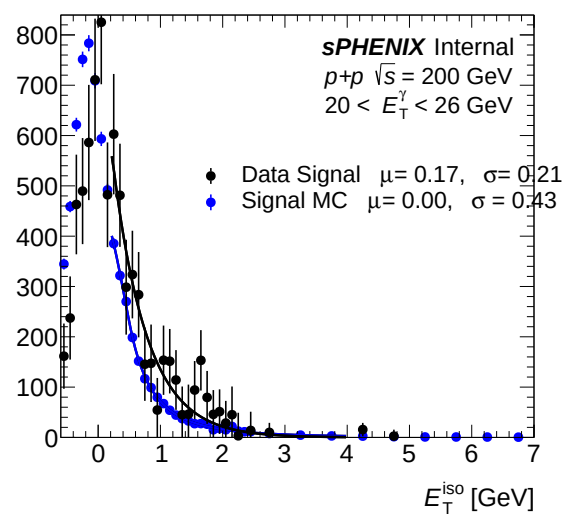

(a) IsoET template (pT bins 6–8) — bdt_nom



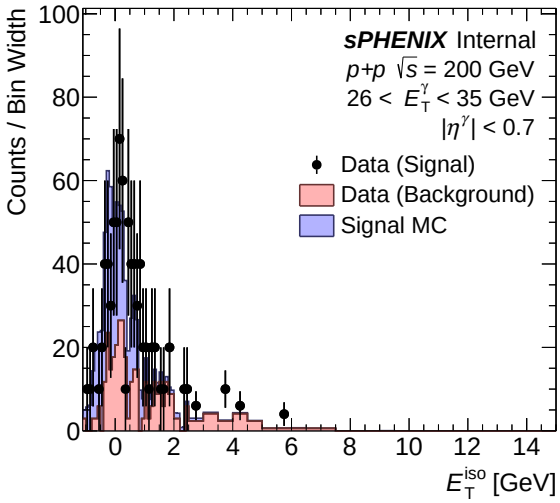

(b) IsoET template (pT bins 6–8) — bdt_tightbdt50



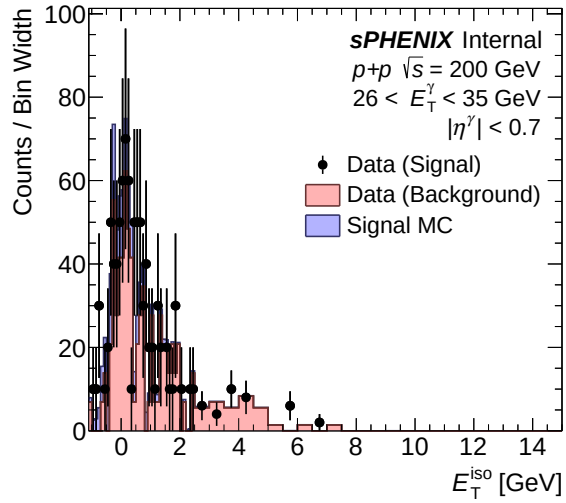

(a) IsoET fit template (pT bins 6–8) — bdt_nom



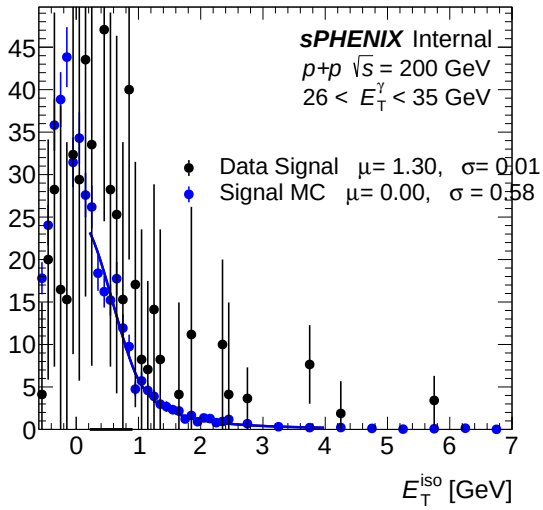

(b) IsoET fit template (pT bins 6–8) — bdt_tightbdt50



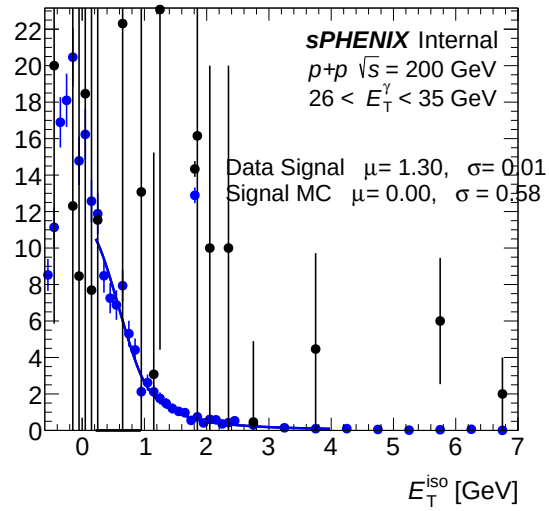

(a) IsoET template (pT bins 9–9) — bdt_nom




(b) IsoET template (pT bins 9–9) — bdt_tightbdt50

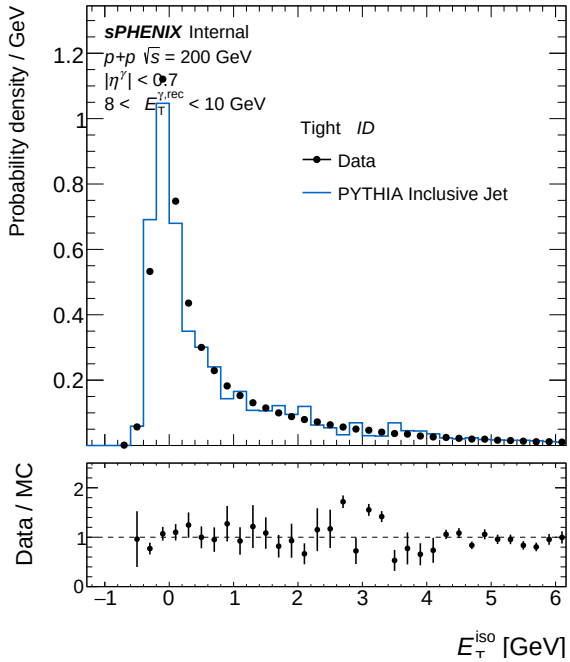



(a) IsoET fit template (pT bins 9–9) — bdt_nom

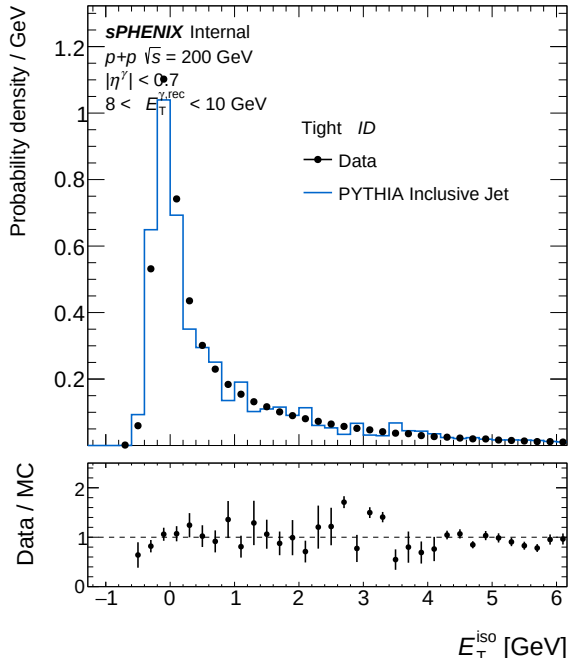



(b) IsoET fit template (pT bins 9–9) — bdt_tightbdt50

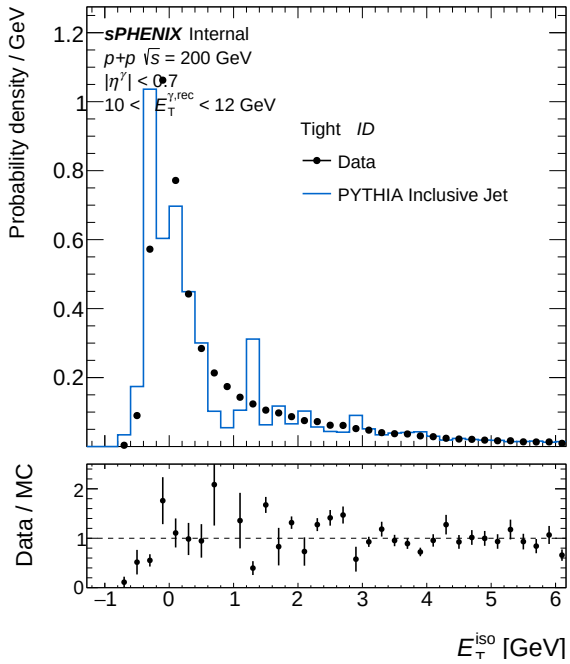
## 1.7 Isolation ET



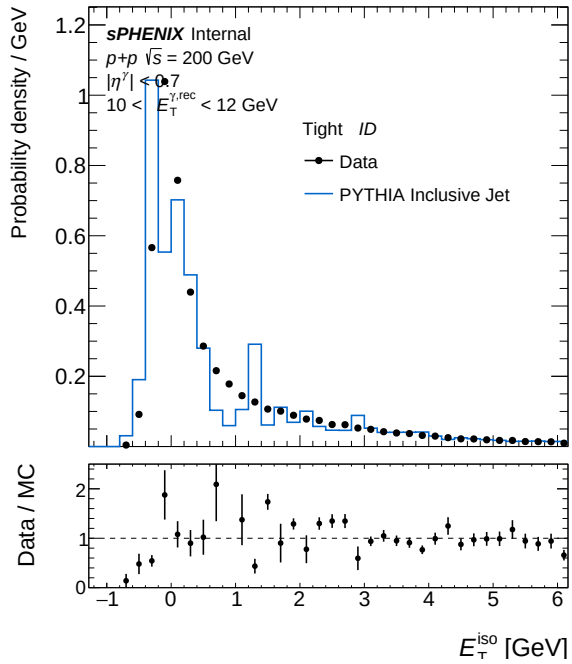

(a) IsoET Tight, pT bin 0 — bdt_nom



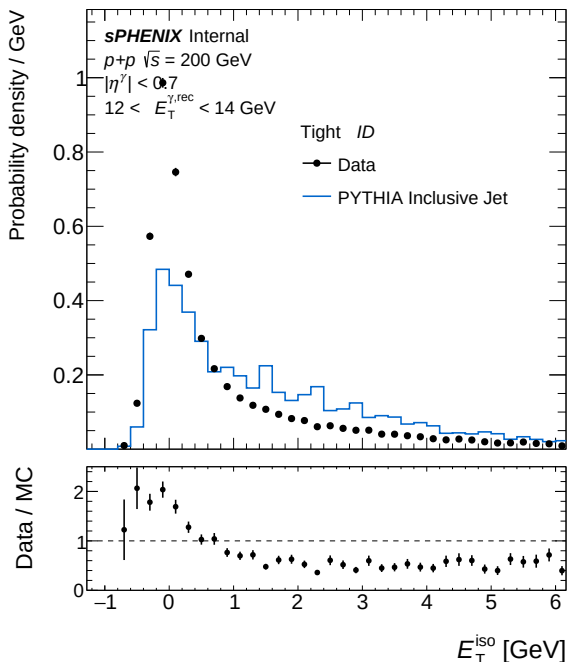

(b) IsoET Tight, pT bin 0 — bdt_tightbdt50



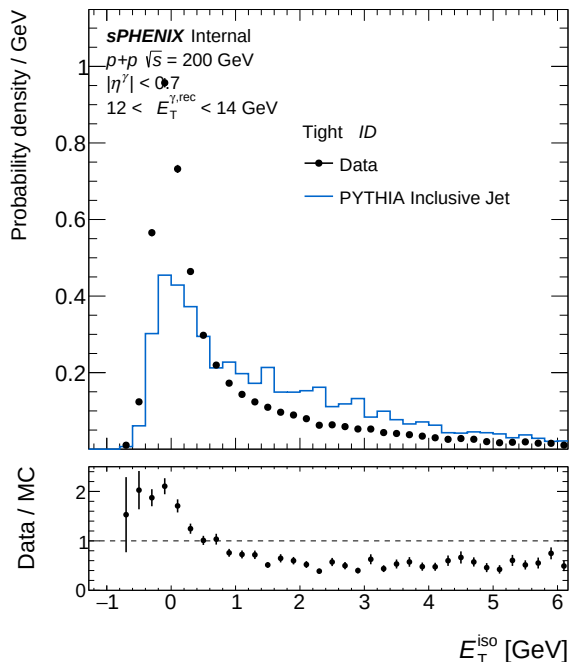

(a) IsoET Tight, pT bin 1 — bdt_nom



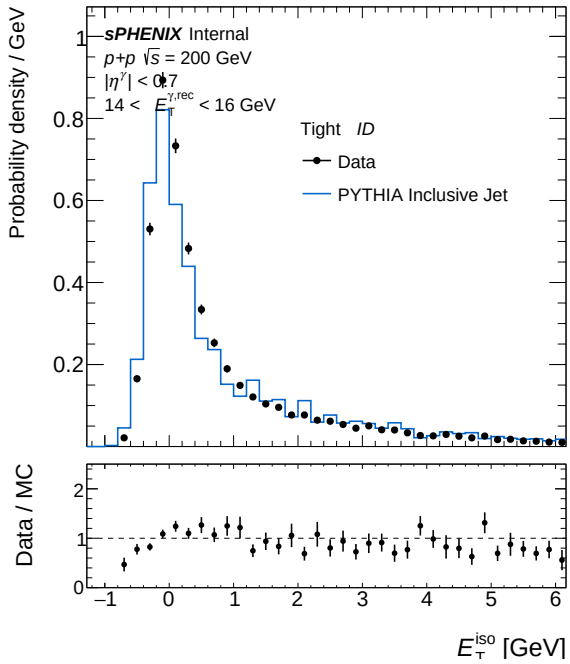

(b) IsoET Tight, pT bin 1 — bdt_tightbdt50



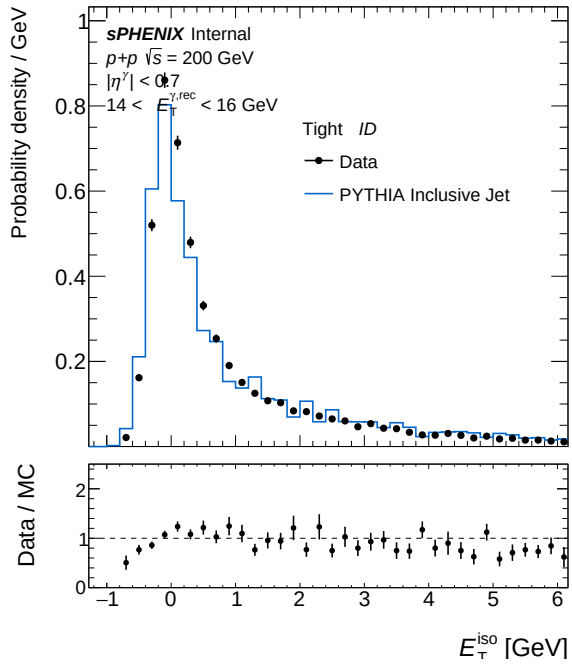

(a) IsoET Tight, pT bin 2 — bdt_nom



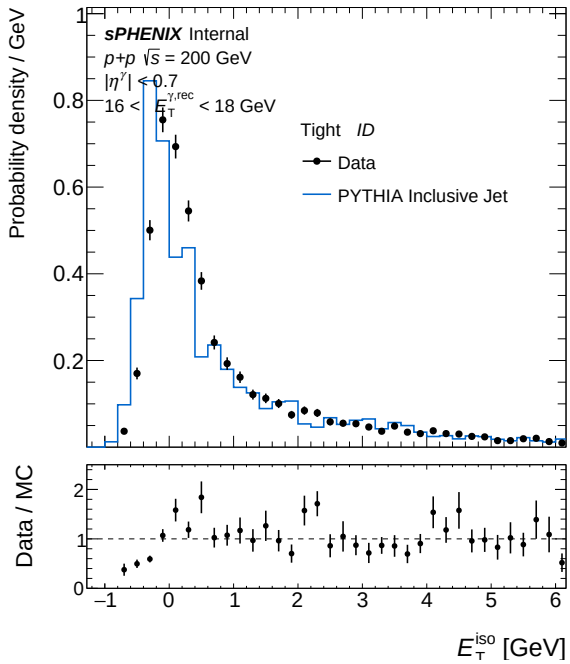

(b) IsoET Tight, pT bin 2 — bdt_tightbdt50



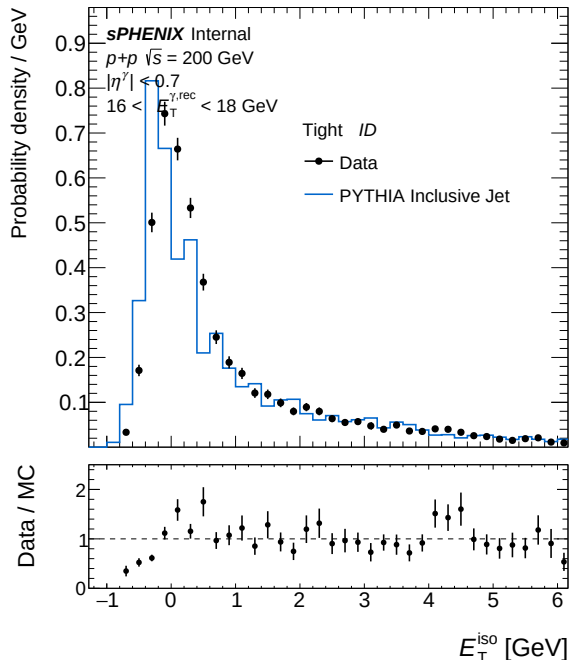

(a) IsoET Tight, pT bin 3 — bdt_nom



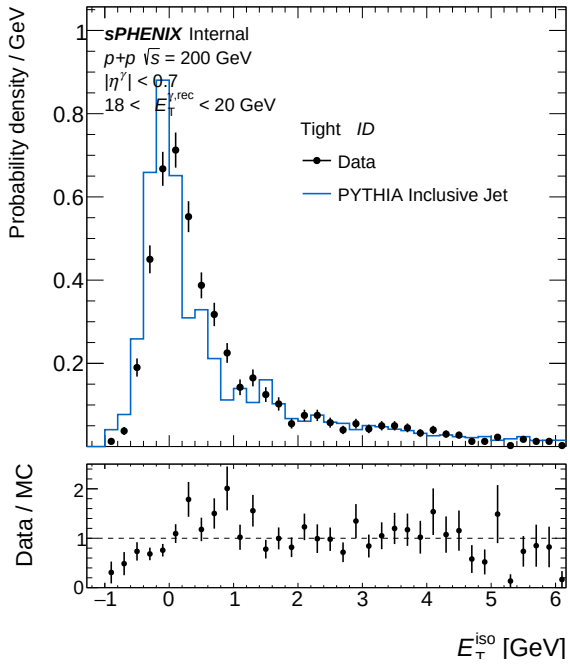

(b) IsoET Tight, pT bin 3 — bdt_tightbdt50



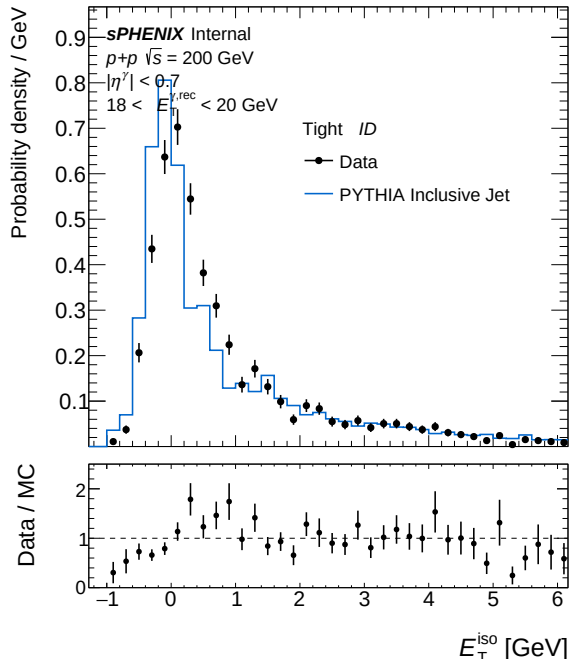

(a) IsoET Tight, pT bin 4 — bdt_nom



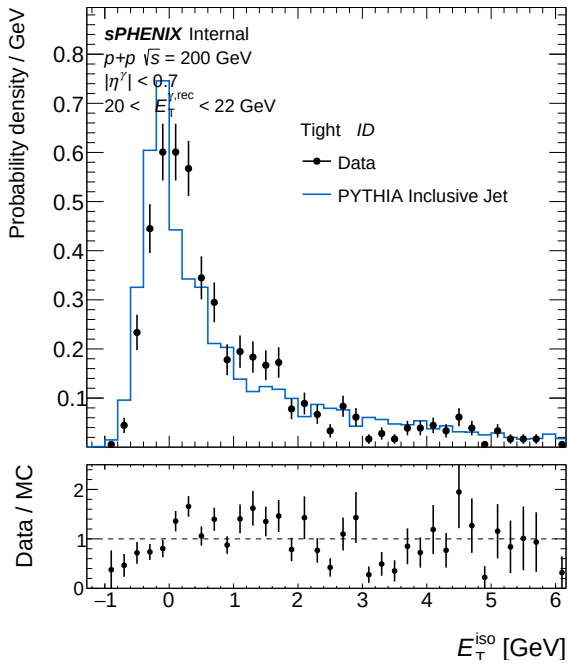

(b) IsoET Tight, pT bin 4 — bdt_tightbdt50



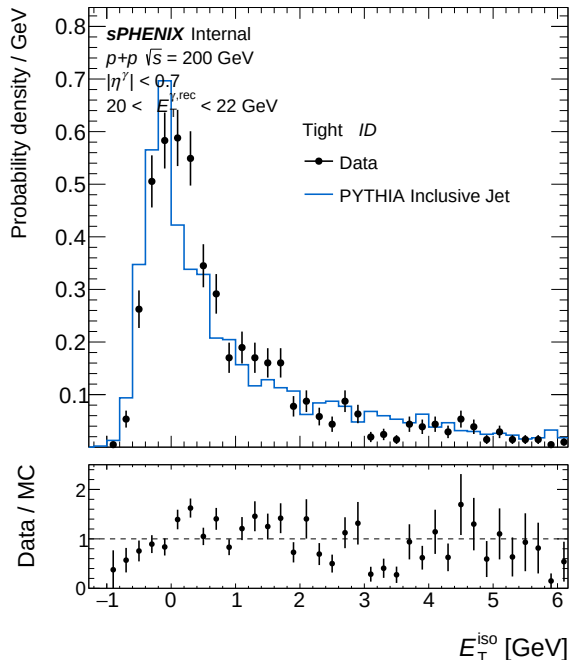

(a) IsoET Tight, pT bin 5 — bdt_nom



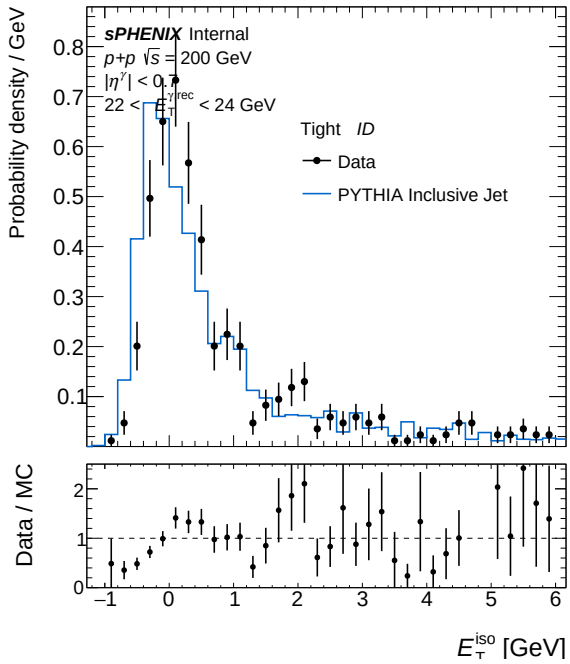

(b) IsoET Tight, pT bin 5 — bdt_tightbdt50



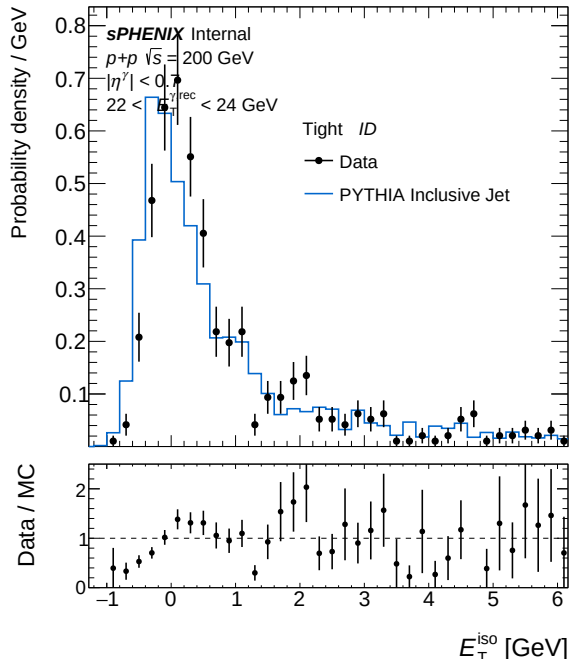

(a) IsoET Tight, pT bin 6 — bdt_nom



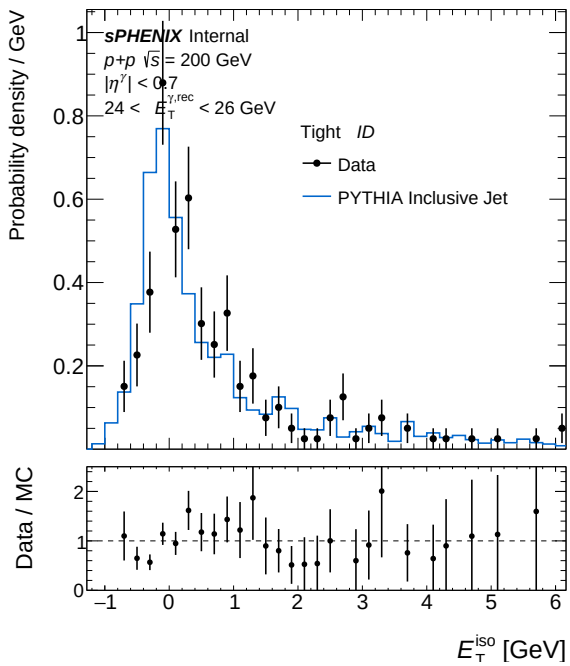

(b) IsoET Tight, pT bin 6 — bdt_tightbdt50



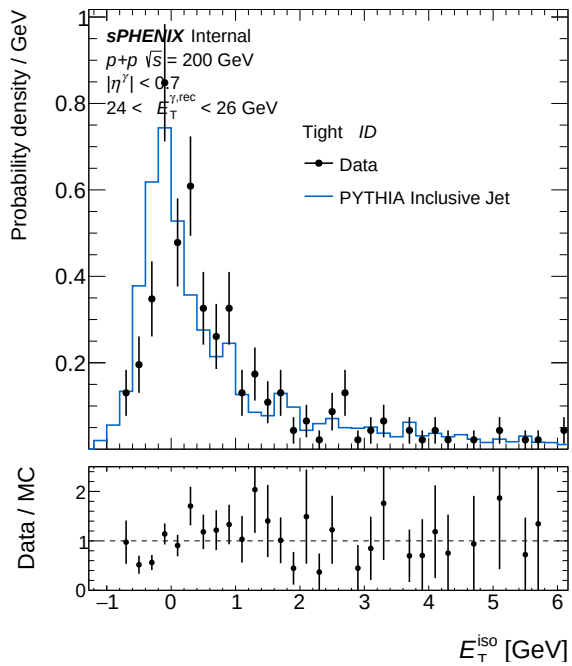

(a) IsoET Tight, pT bin 7 — bdt_nom



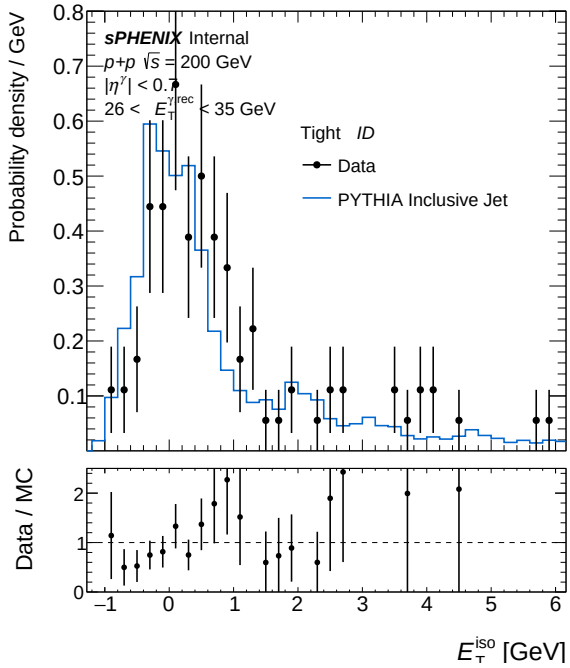

(b) IsoET Tight, pT bin 7 — bdt_tightbdt50



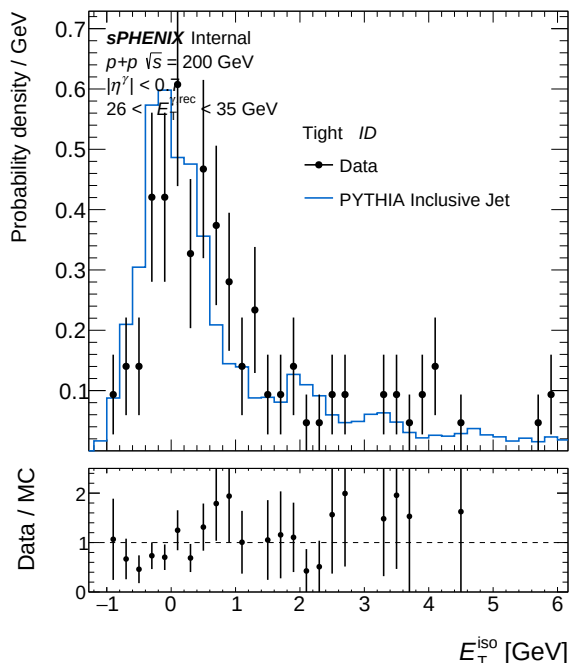

(a) IsoET Tight, pT bin 8 — bdt_nom




(b) IsoET Tight, pT bin 8 — bdt_tightbdt50




(a) IsoET Tight, pT bin 9 — bdt_nom



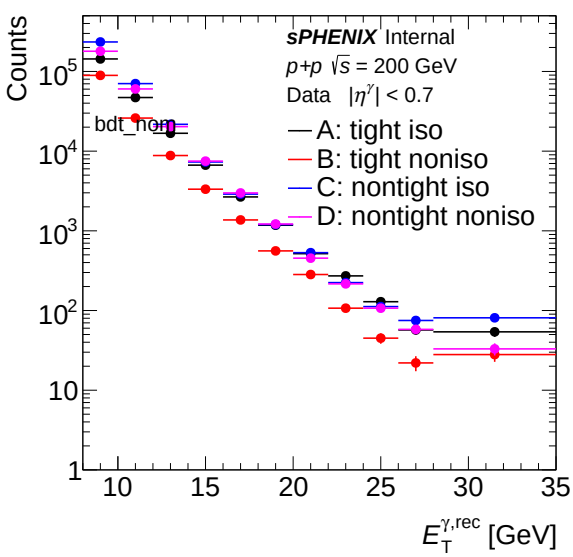

(b) IsoET Tight, pT bin 9 — bdt_tightbdt50

# 2 bdt_nom vs bdt_npb03

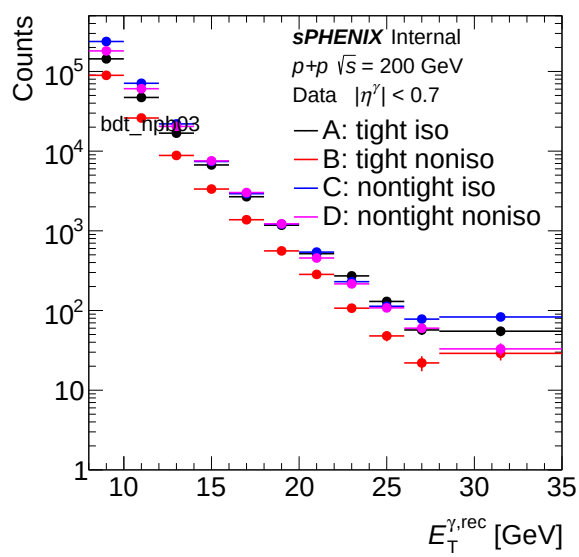
## Config diff (A → B)

```
--- config_bdt_nom.yaml
+++ config_bdt_npb03.yaml
@@ -15,5 +15,5 @@
   response_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/MC_response"
   data_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/data_histo"
-  var_type: "bdt_nom"
+  var_type: "bdt_npb03"
   final_outfile: "/sphenix/user/shuhangli/ppg12/efficiencytool/results/Photon_final"

@@ -42,6 +42,6 @@
   eff_dR: 0.1
   trigger_used: 30
-  pT_bins: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30]
-  pT_bins_truth: [7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 35]
+  pT_bins: [8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 35]
+  pT_bins_truth: [7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 35, 40]
   iso_threshold: 1
   iso_hcalonly: 0
@@ -191,5 +191,5 @@

     npb_cut_on: 1
-    npb_score_cut: 0.5
+    npb_score_cut: 0.3

     prob_max: 1.0
```

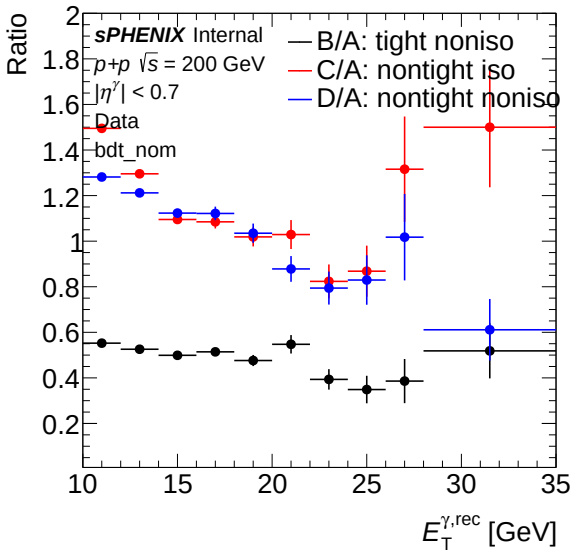
### 2.1 Sideband (Data)



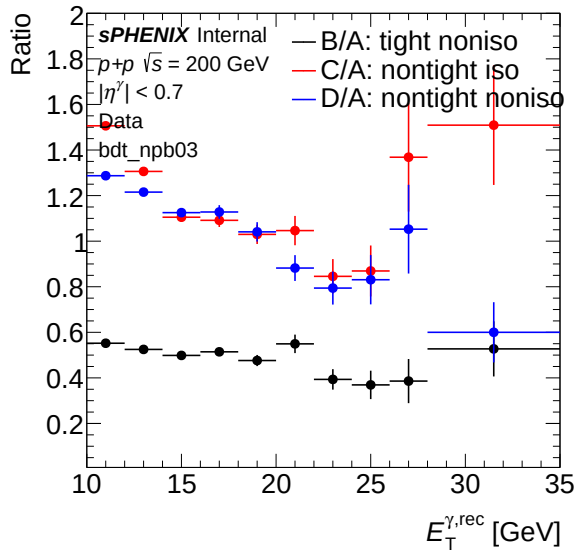

(a) Sideband ET spectrum (data) — bdt_nom



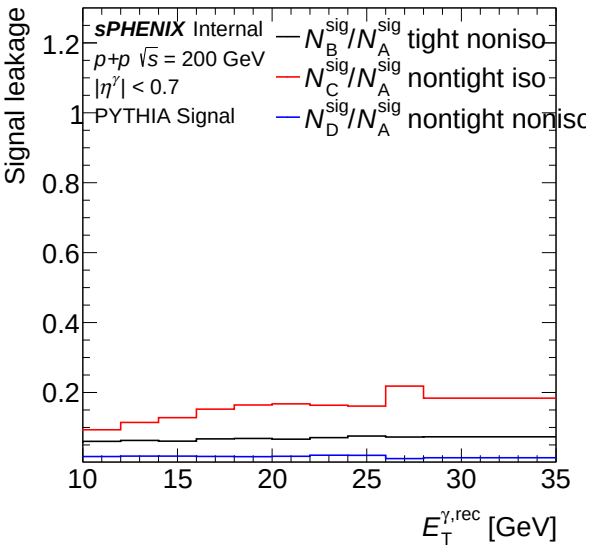

(b) Sideband ET spectrum (data) — bdt_npb03



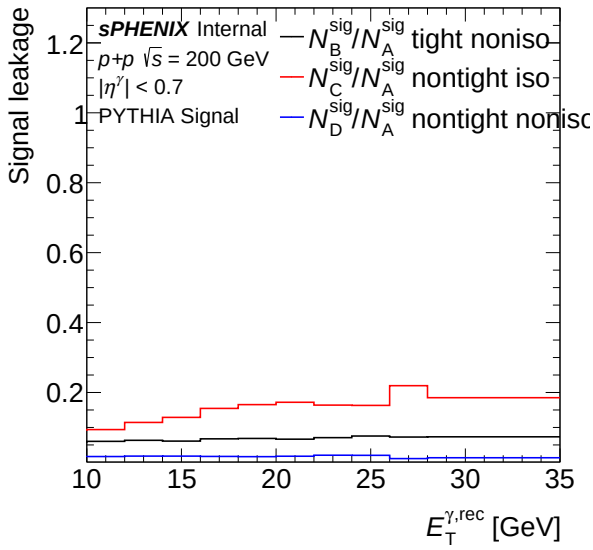

(a) Sideband ET ratio (data) — bdt_nom




(b) Sideband ET ratio (data) — bdt_npb03

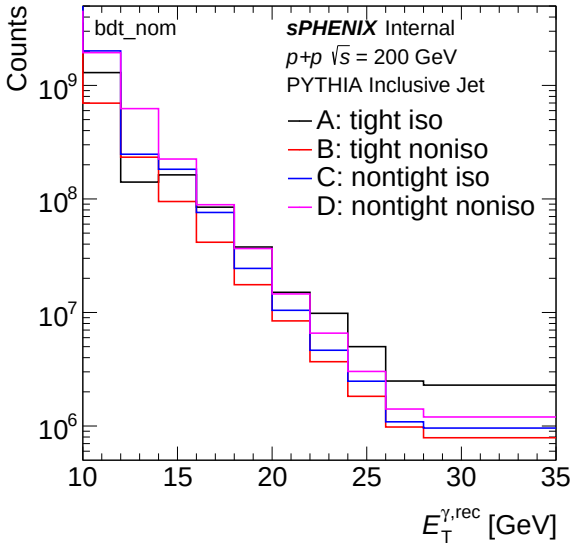



(a) Leakage fraction vs ET (data) — bdt_nom

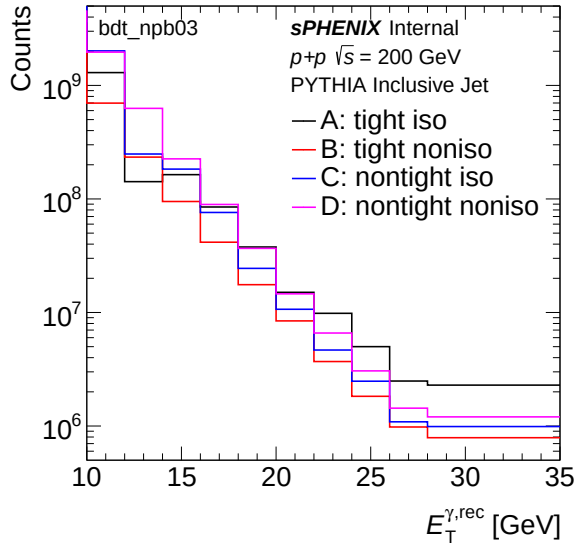



(b) Leakage fraction vs ET (data) — bdt_npb03

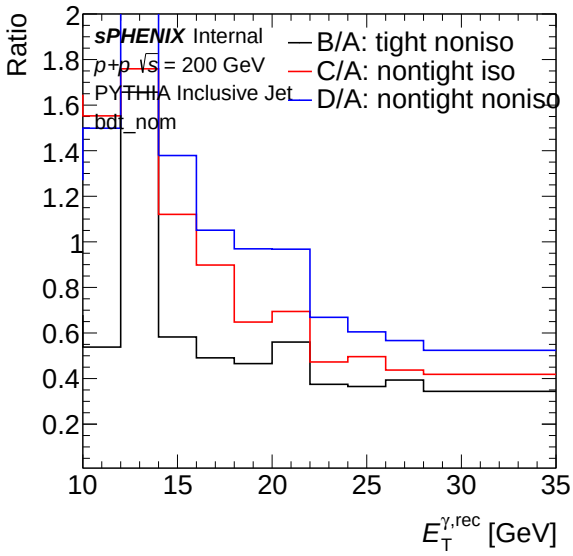
## 2.2 Sideband (MC)



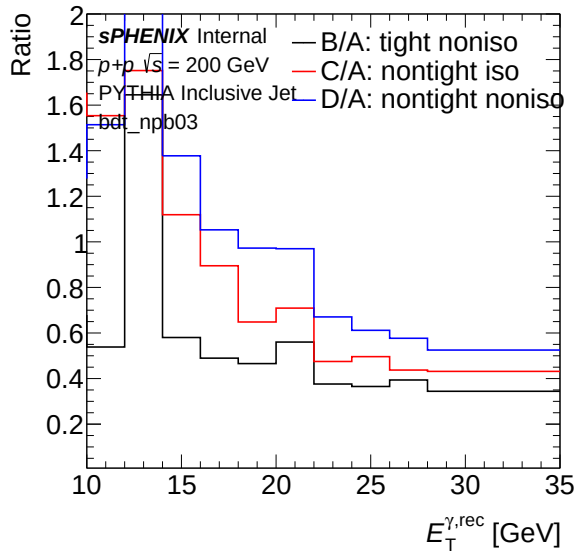

(a) Sideband ET spectrum (MC) — bdt_nom



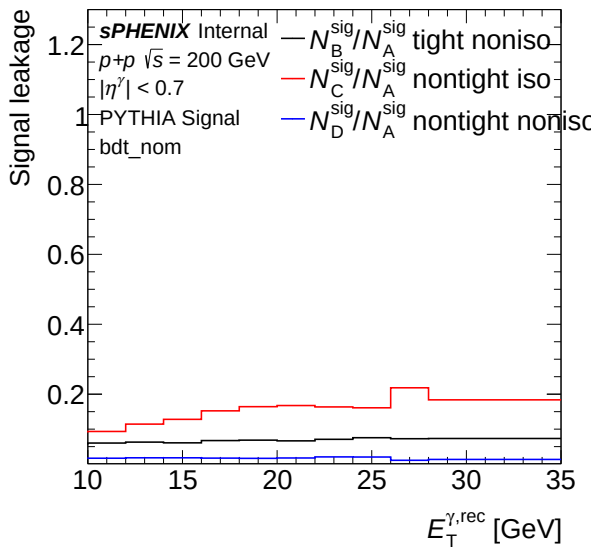

(b) Sideband ET spectrum (MC) — bdt_npb03



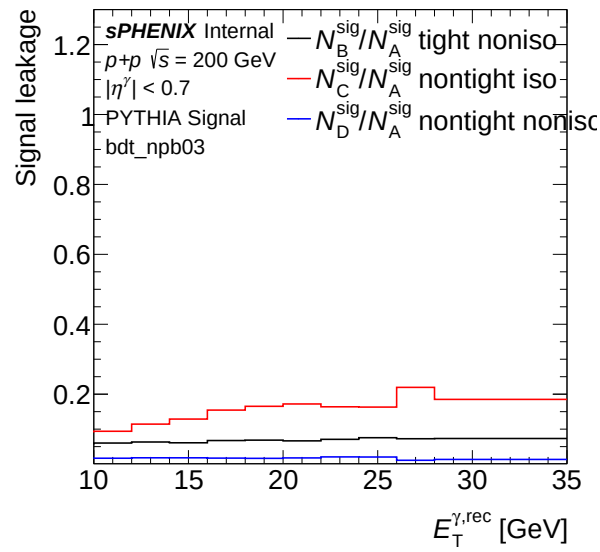

(a) Sideband ET ratio (MC) — bdt_nom




(b) Sideband ET ratio (MC) — bdt_npb03

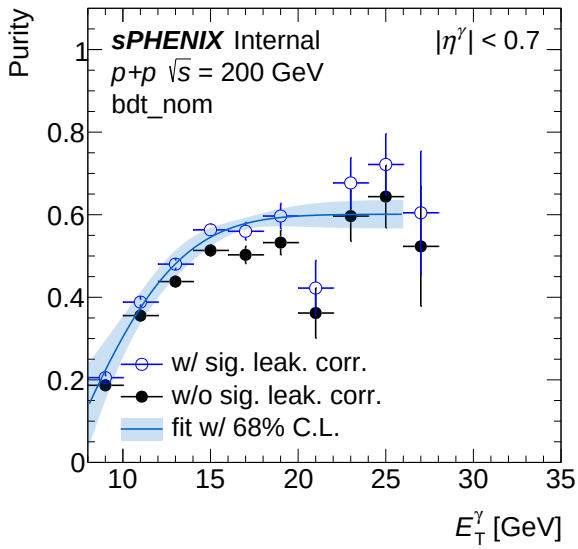



(a) Leakage fraction vs ET (MC) — bdt_nom

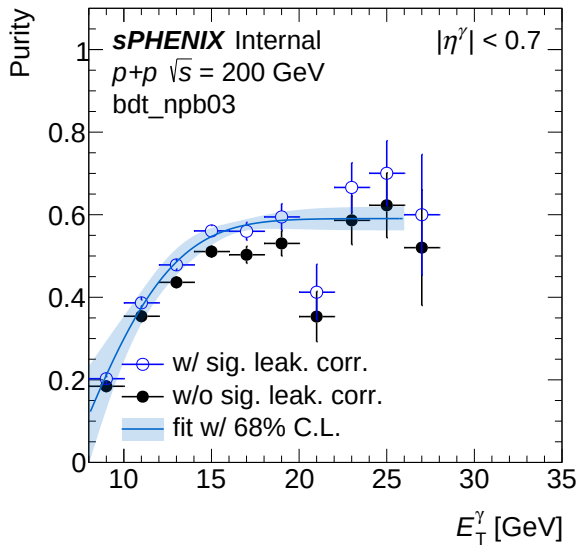



(b) Leakage fraction vs ET (MC) — bdt_npb03

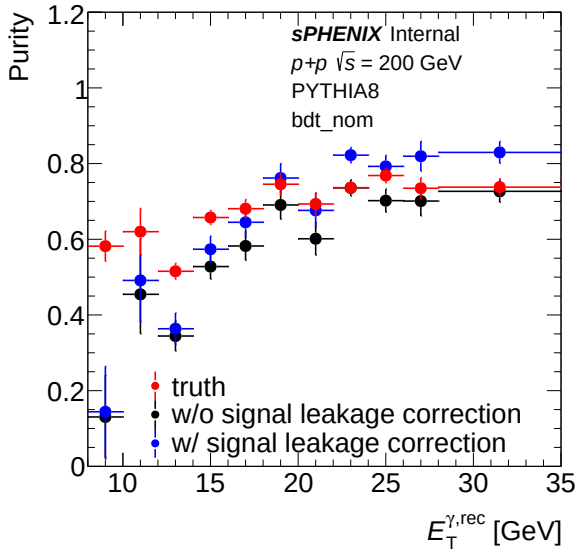
## 2.3 Purity



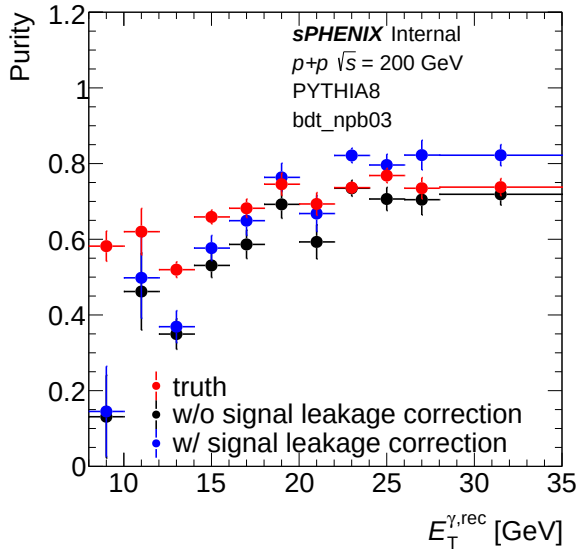

(a) Purity (data) — bdt_nom




(b) Purity (data) — bdt_npb03

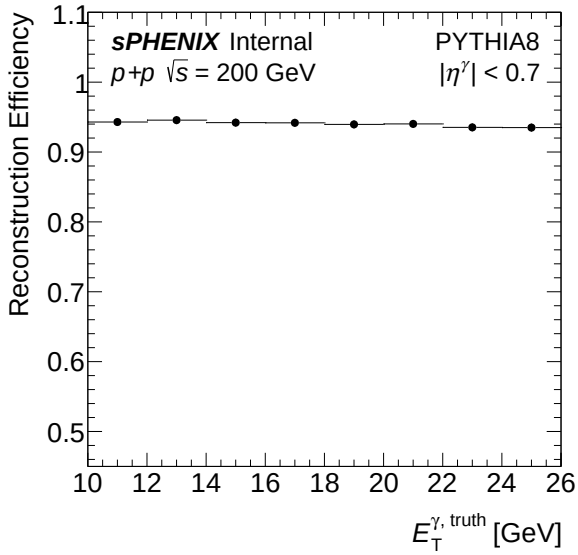



(a) Purity (MC) — bdt_nom

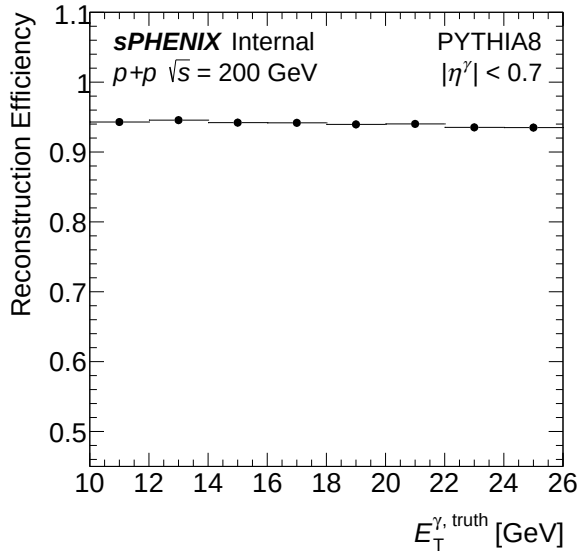



(b) Purity (MC) — bdt_npb03

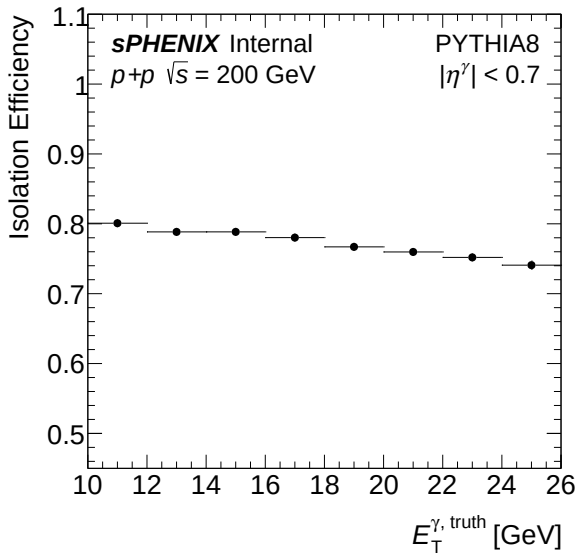
## 2.4 Efficiency



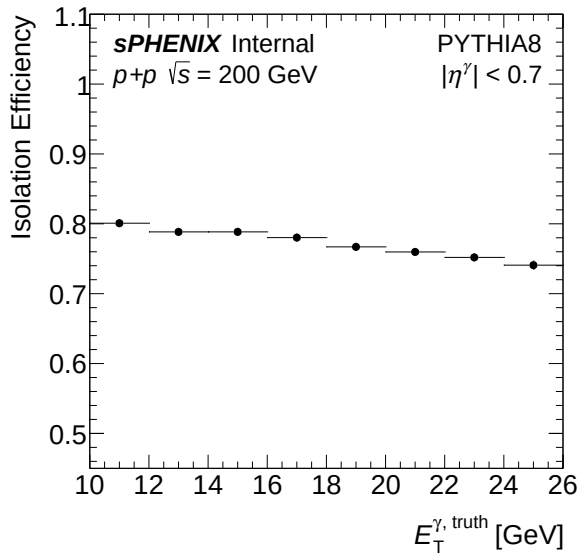

(a) Reconstruction efficiency — bdt_nom



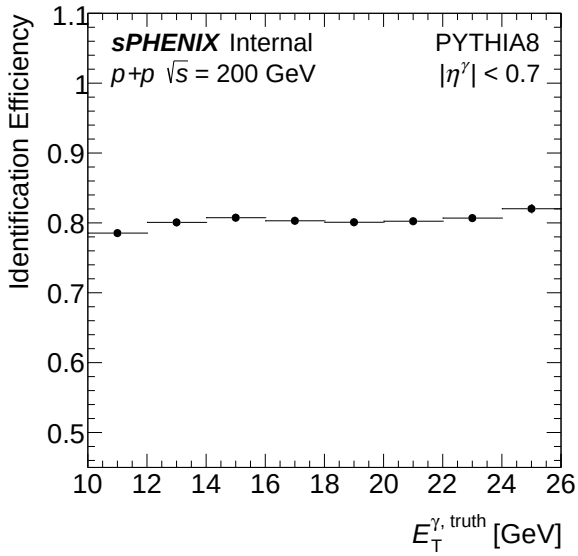

(b) Reconstruction efficiency — bdt_npb03



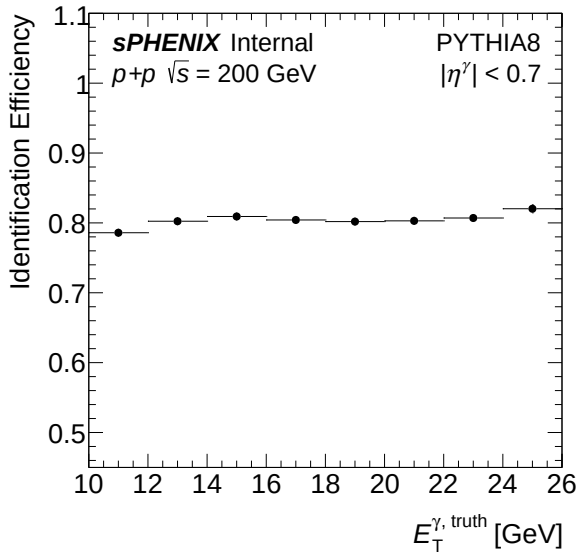

(a) Isolation efficiency — bdt_nom



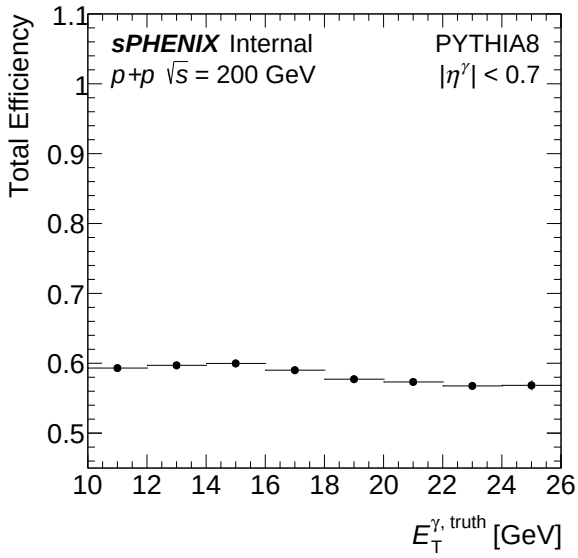

(b) Isolation efficiency — bdt_npb03



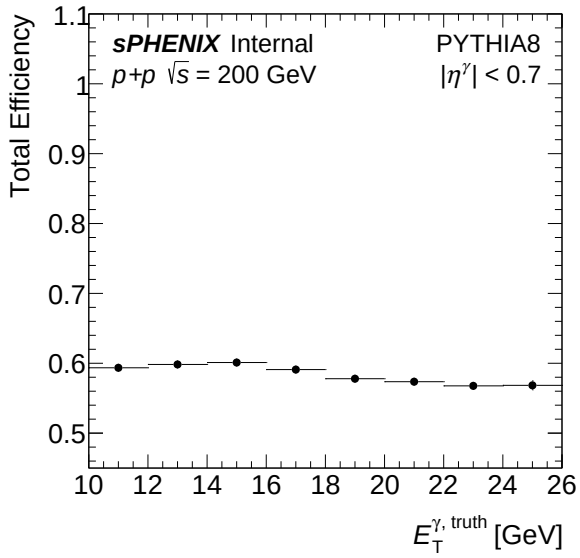

(a) Identification efficiency — bdt_nom



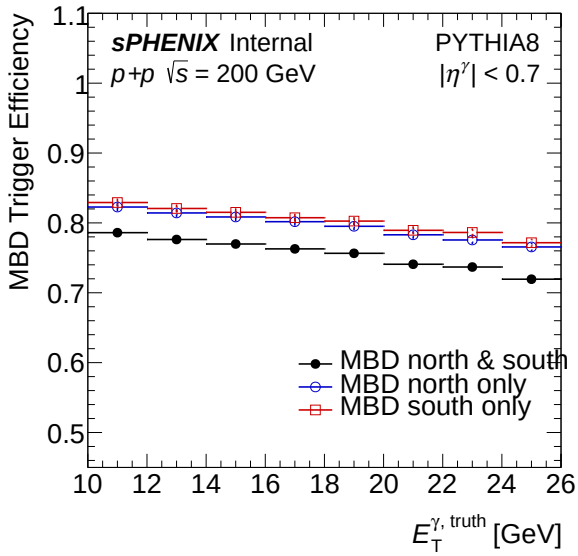

(b) Identification efficiency — bdt_npb03



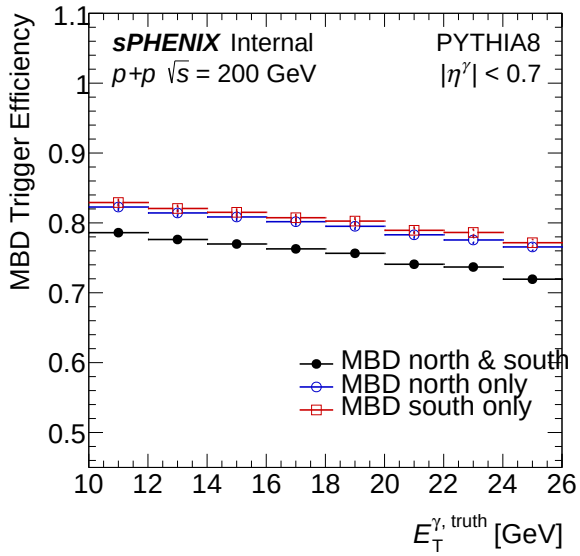

(a) Total efficiency — bdt_nom



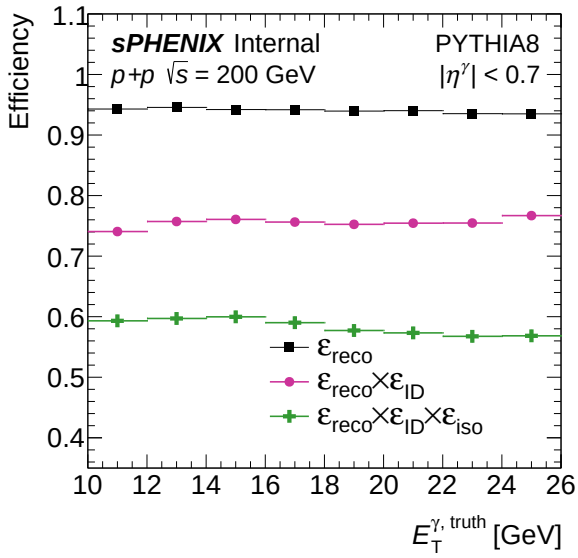

(b) Total efficiency — bdt_npb03



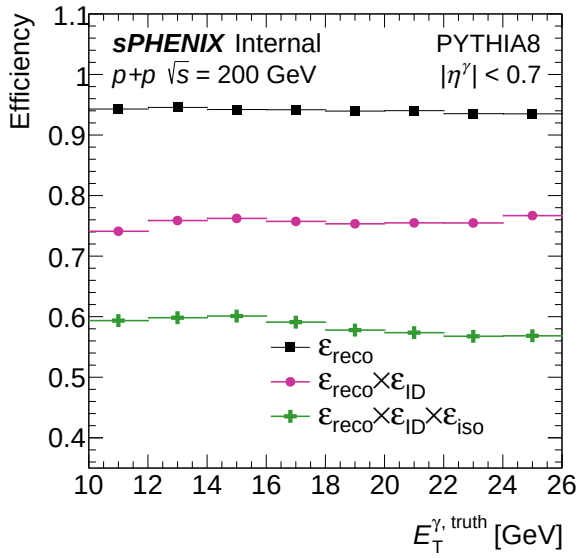

(a) MBD trigger efficiency — bdt_nom




(b) MBD trigger efficiency — bdt_npb03

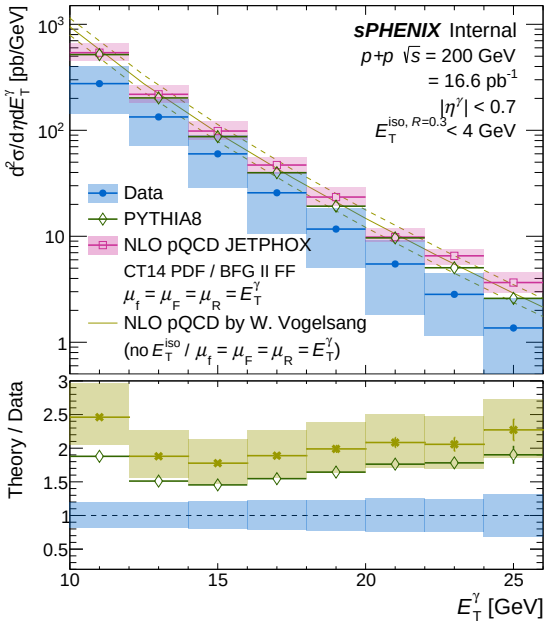



(a) Combined photon efficiency — bdt_nom

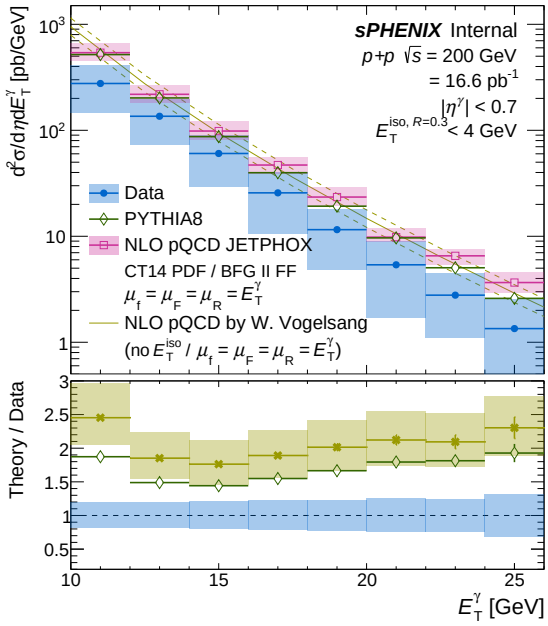



(b) Combined photon efficiency — bdt_npb03

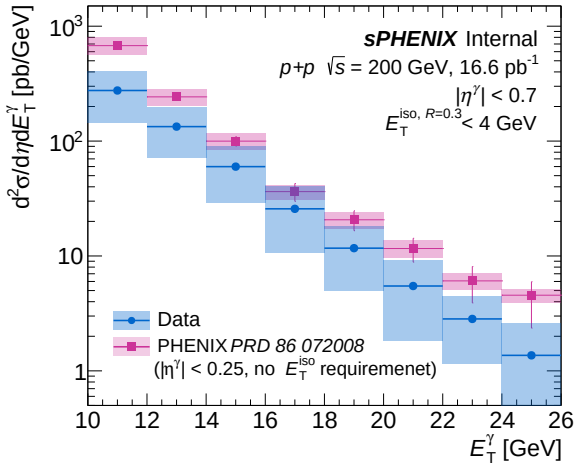
## 2.5 Final result



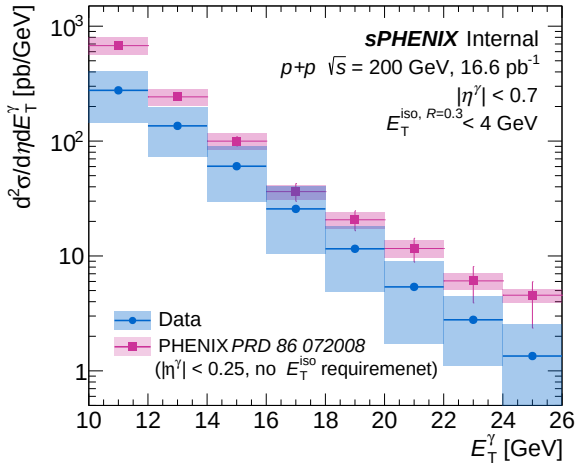

(a) Final cross section result — bdt_nom



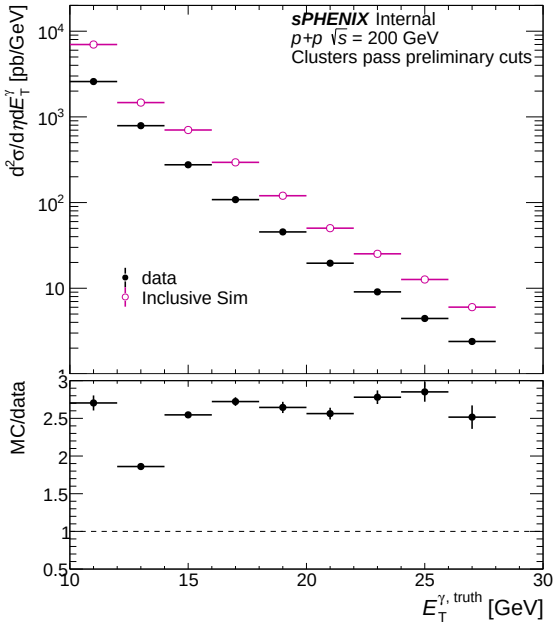

(b) Final cross section result — bdt_npb03



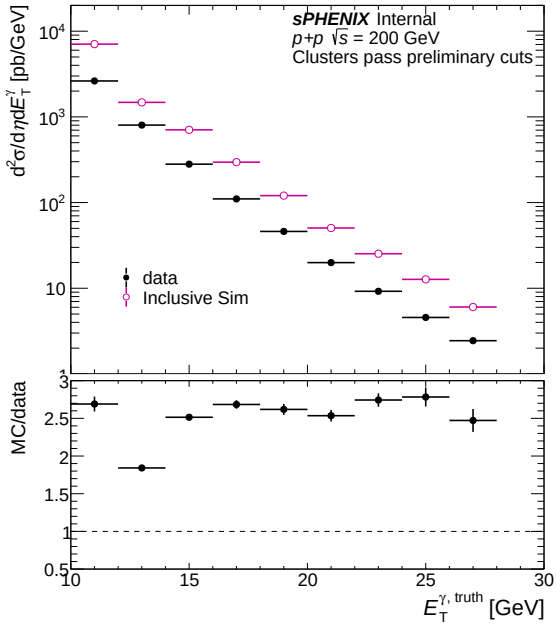

(a) Comparison with PHENIX — bdt_nom



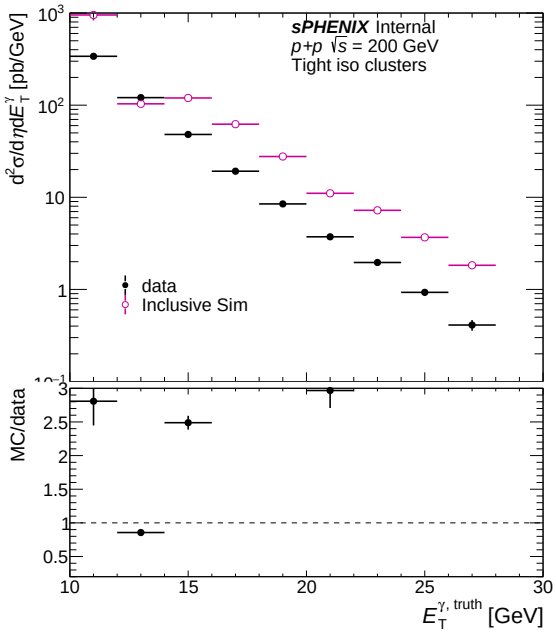

(b) Comparison with PHENIX — bdt_npb03



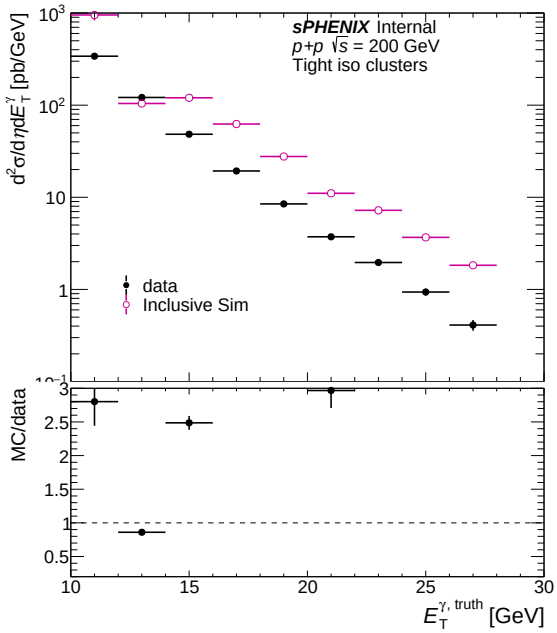

(a) Common cluster yield (data vs MC) — bdt_nom



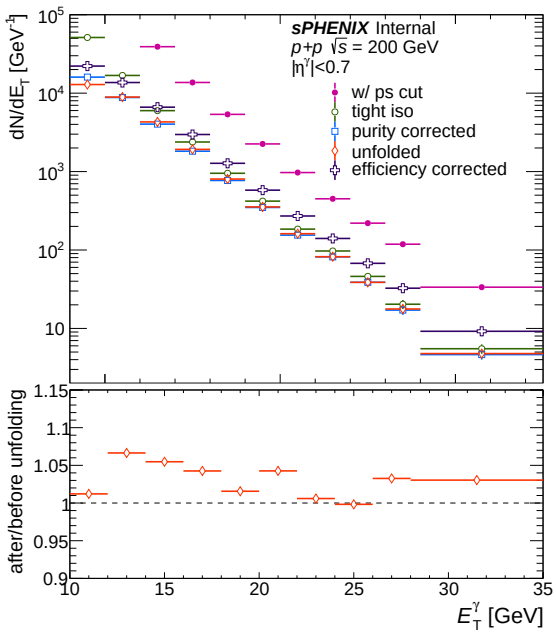

(b) Common cluster yield (data vs MC) — bdt_npb03



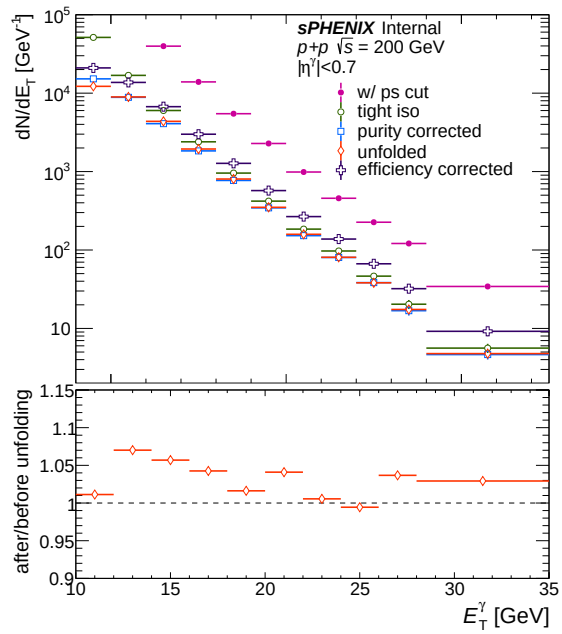

(a) Tight iso cluster yield (data vs MC) — bdt_nom




(b) Tight iso cluster yield (data vs MC) — bdt_npb03

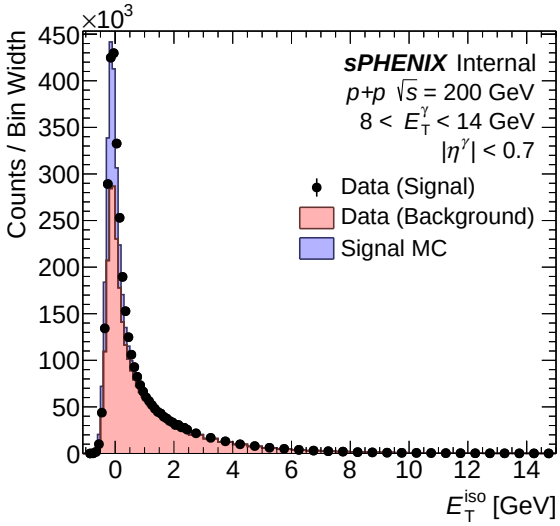



(a) All correction steps — bdt_nom

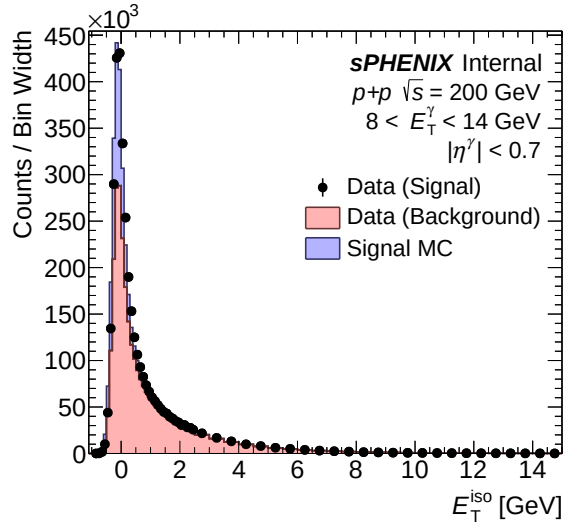



(b) All correction steps — bdt_npb03

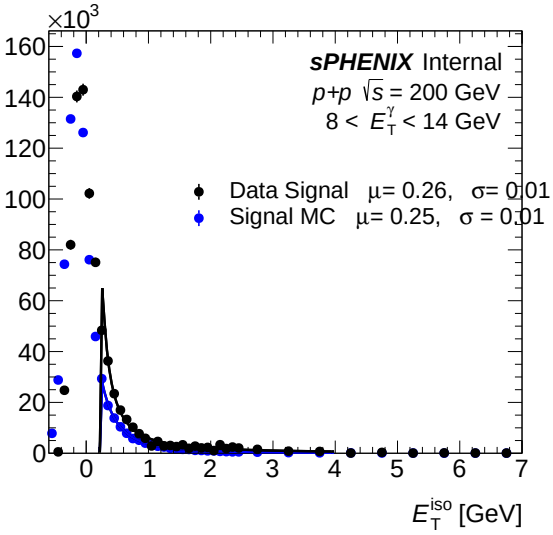
## 2.6 Isolation fit (CONF)



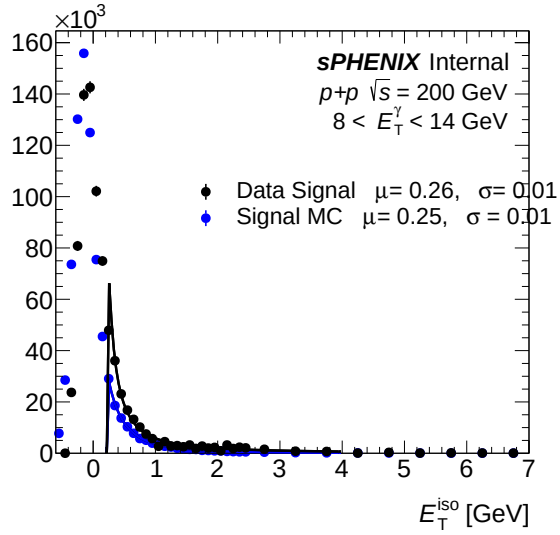

(a) IsoET template (pT bins 0–2) — bdt_nom



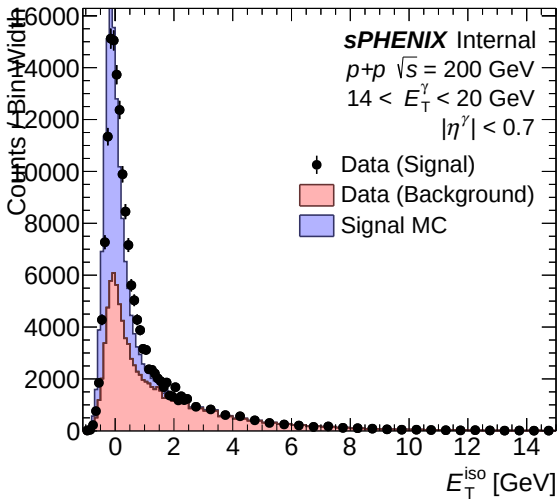

(b) IsoET template (pT bins 0–2) — bdt_npb03



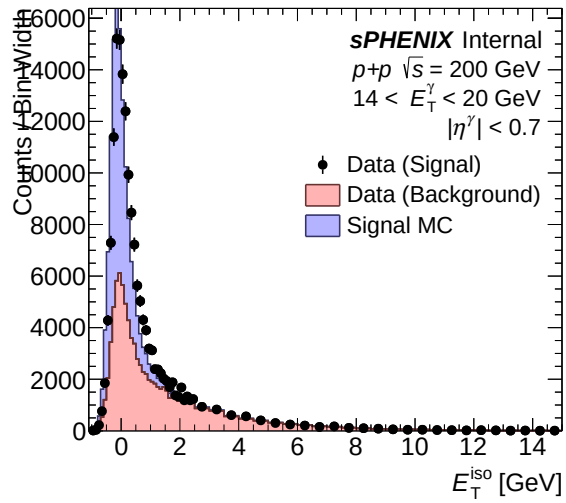

(a) IsoET fit template (pT bins 0–2) — bdt_nom



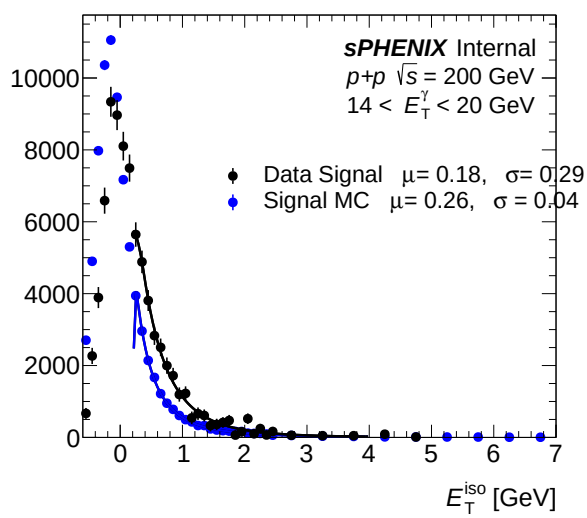

(b) IsoET fit template (pT bins 0–2) — bdt_npb03



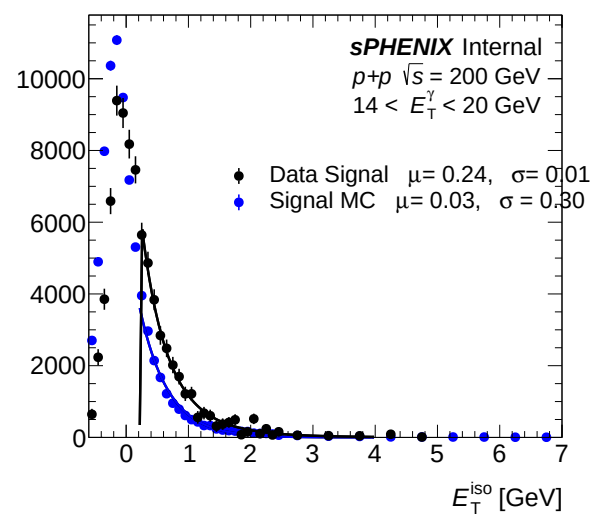

(a) IsoET template (pT bins 3–5) — bdt_nom



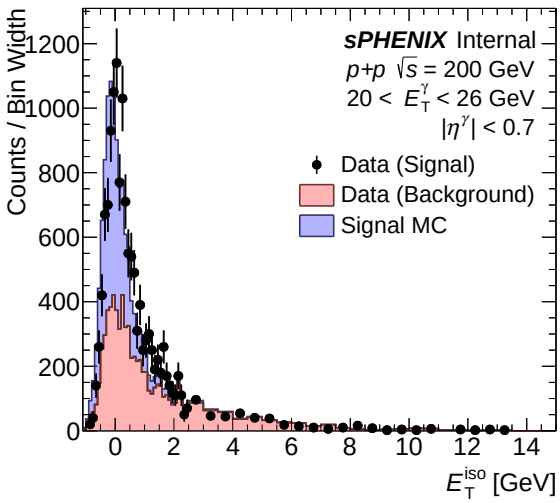

(b) IsoET template (pT bins 3–5) — bdt_npb03



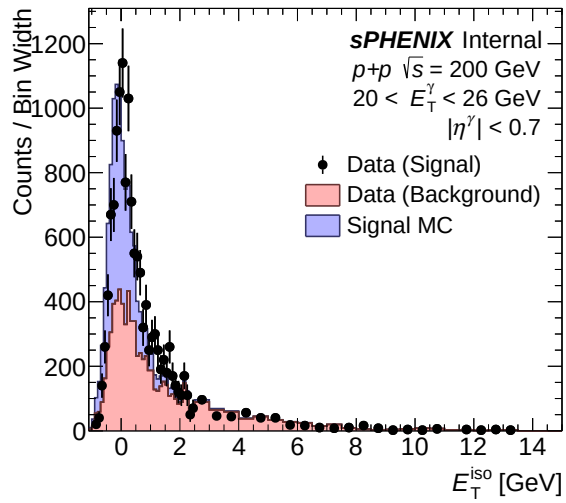

(a) IsoET fit template (pT bins 3–5) — bdt_nom



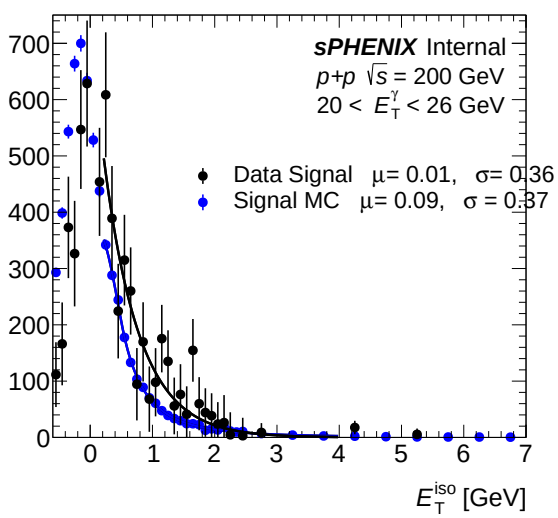

(b) IsoET fit template (pT bins 3–5) — bdt_npb03



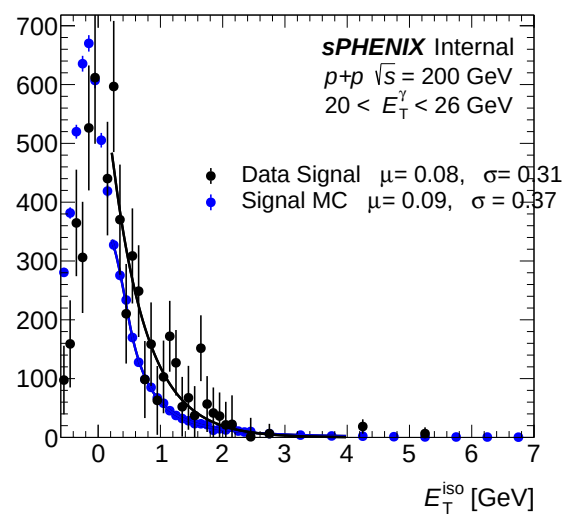

(a) IsoET template (pT bins 6–8) — bdt_nom



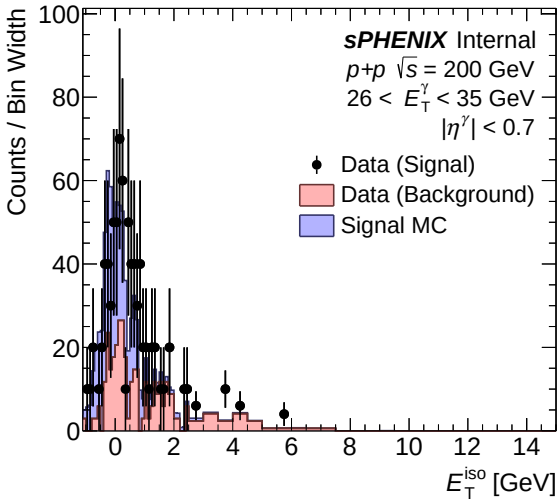

(b) IsoET template (pT bins 6–8) — bdt_npb03



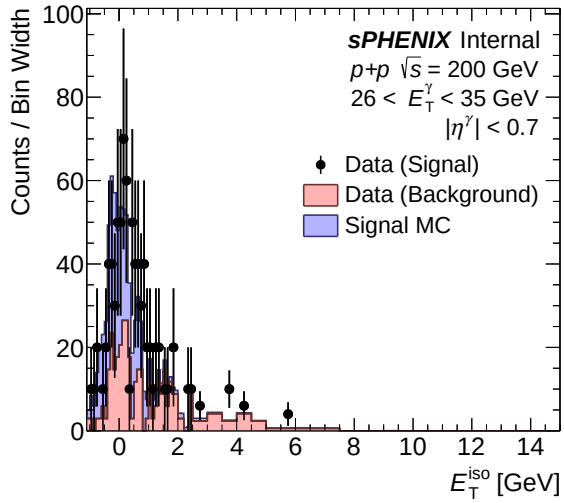

(a) IsoET fit template (pT bins 6–8) — bdt_nom



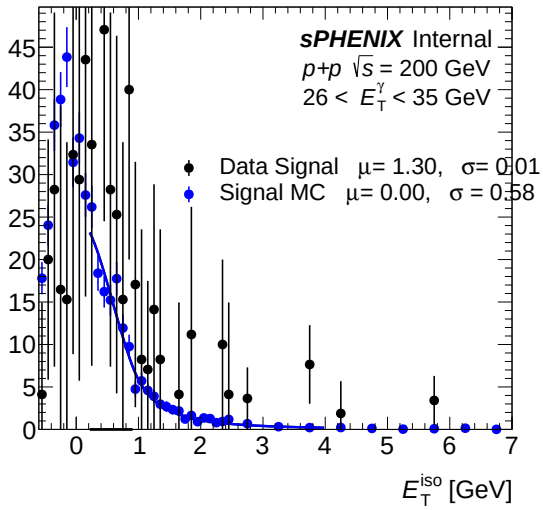

(b) IsoET fit template (pT bins 6–8) — bdt_npb03



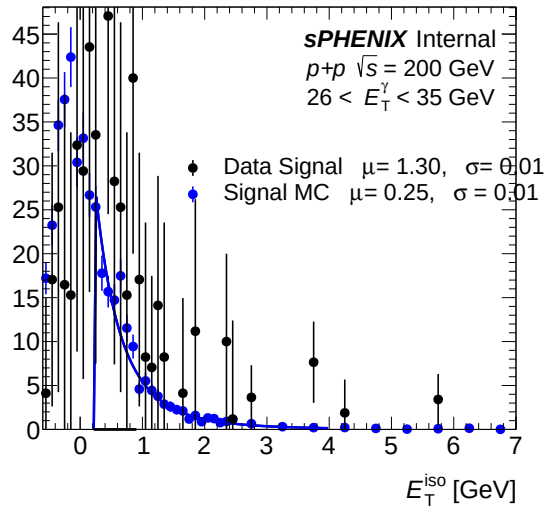

(a) IsoET template (pT bins 9–9) — bdt_nom




(b) IsoET template (pT bins 9–9) — bdt_npb03

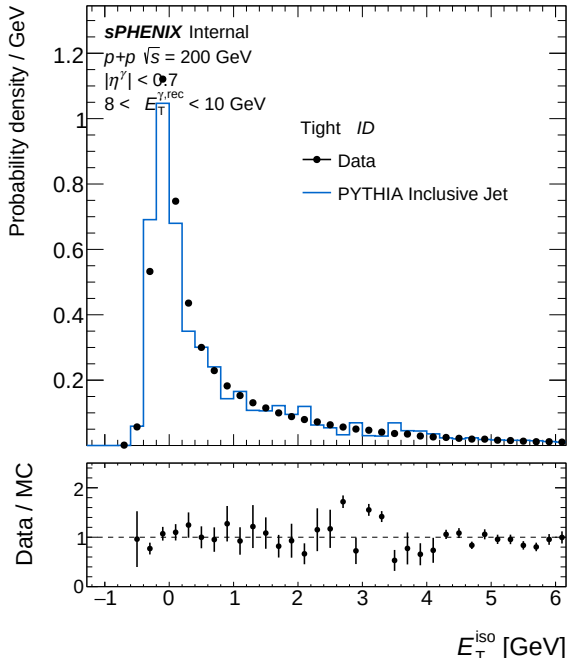



(a) IsoET fit template (pT bins 9–9) — bdt_nom

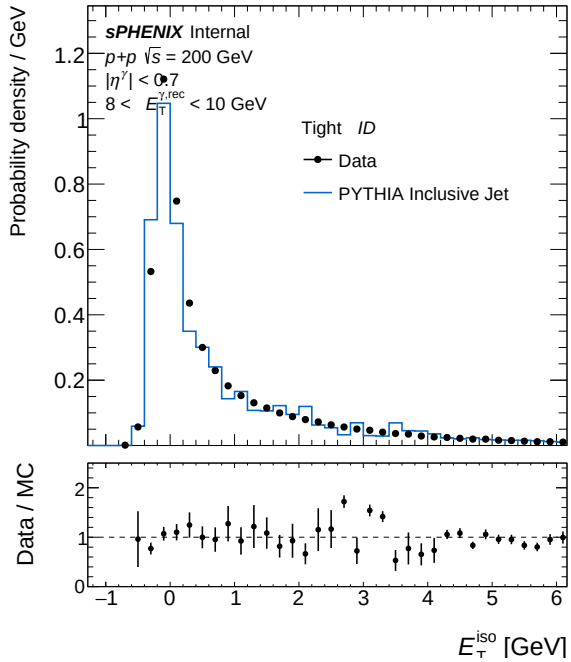



(b) IsoET fit template (pT bins 9–9) — bdt_npb03

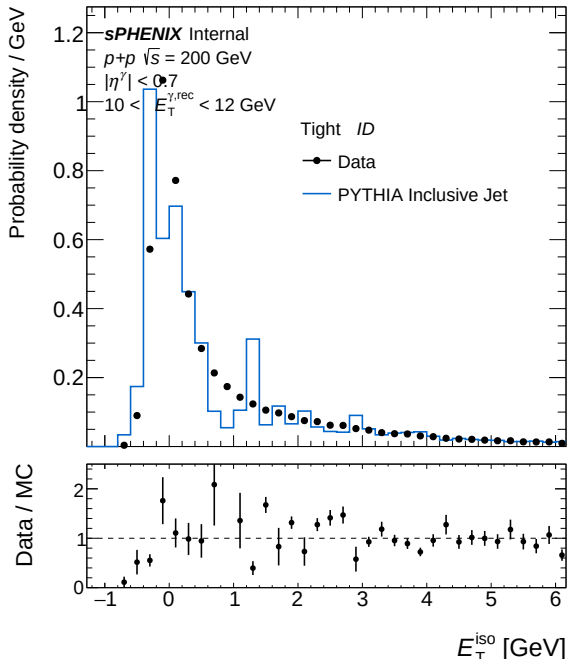
## 2.7 Isolation ET



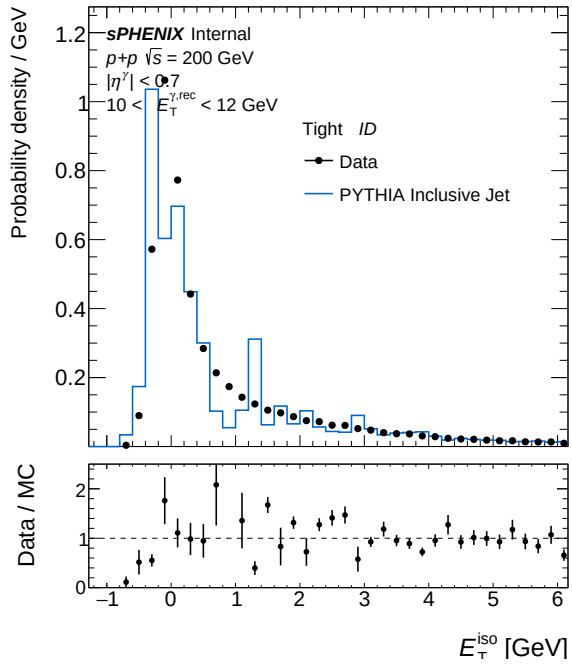

(a) IsoET Tight, pT bin 0 — bdt_nom



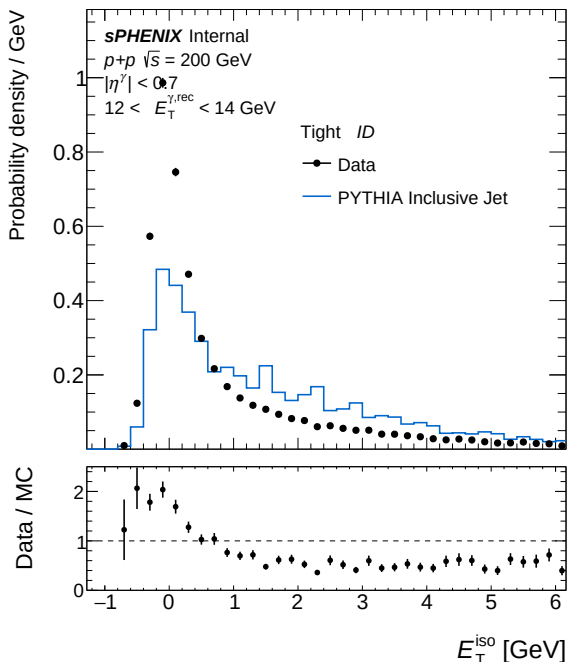

(b) IsoET Tight, pT bin 0 — bdt_npb03



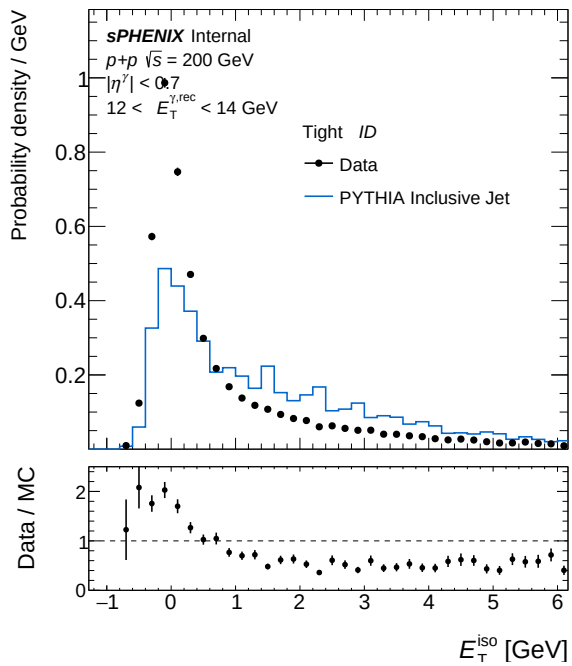

(a) IsoET Tight, pT bin 1 — bdt_nom



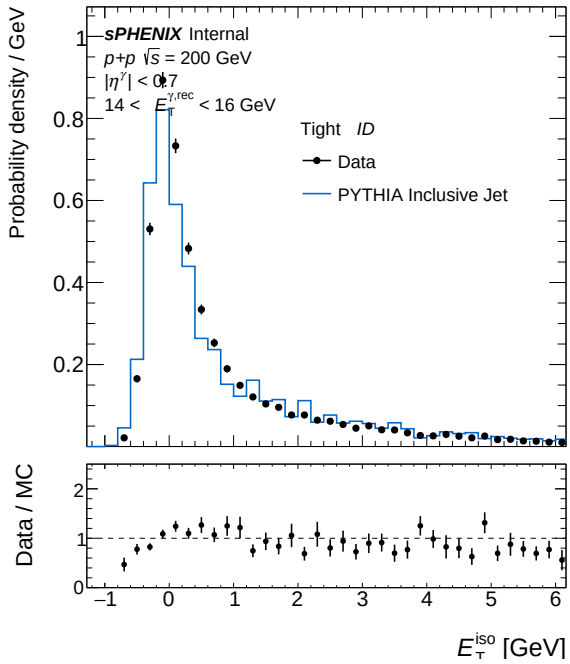

(b) IsoET Tight, pT bin 1 — bdt_npb03



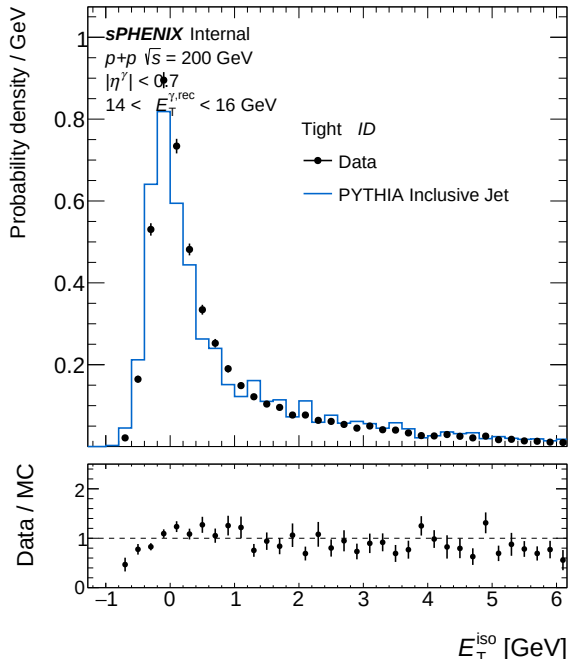

(a) IsoET Tight, pT bin 2 — bdt_nom



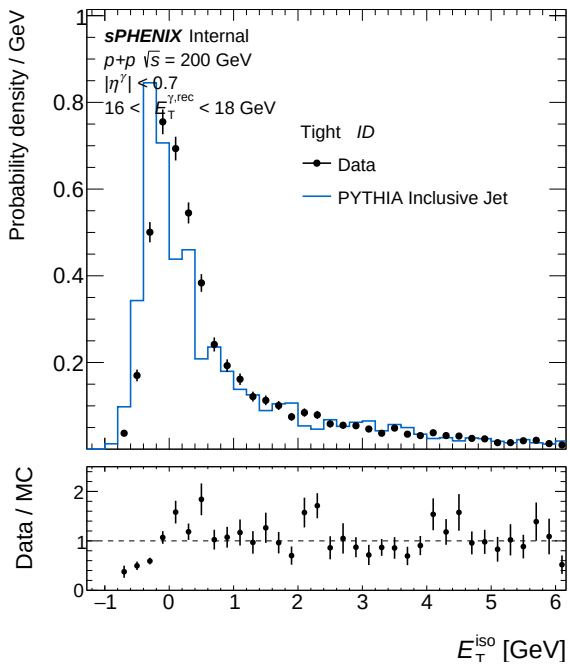

(b) IsoET Tight, pT bin 2 — bdt_npb03



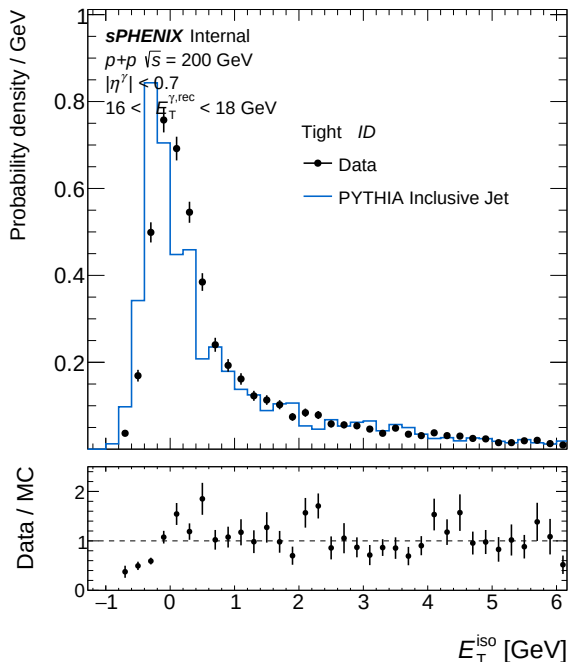

(a) IsoET Tight, pT bin 3 — bdt_nom



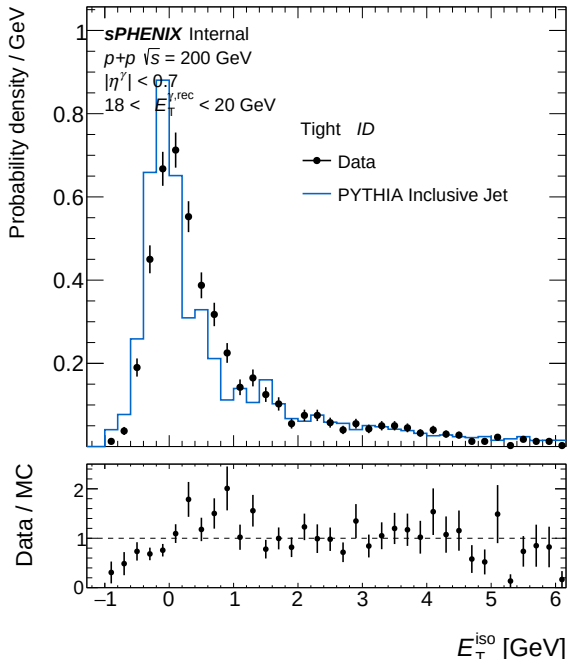

(b) IsoET Tight, pT bin 3 — bdt_npb03



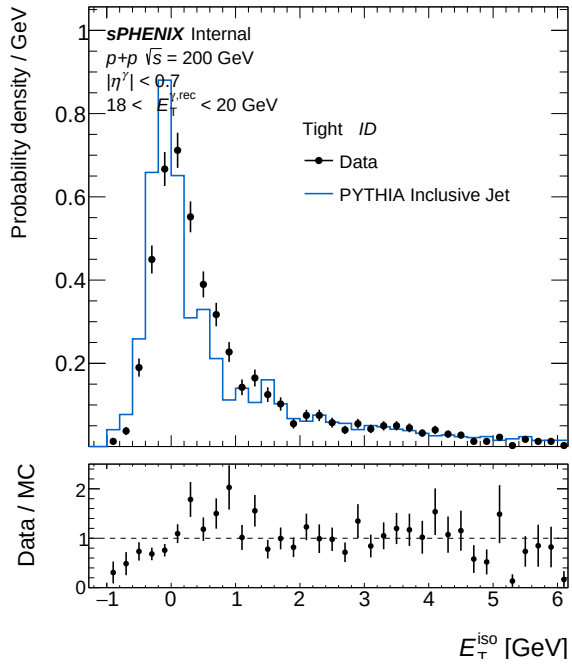

(a) IsoET Tight, pT bin 4 — bdt_nom



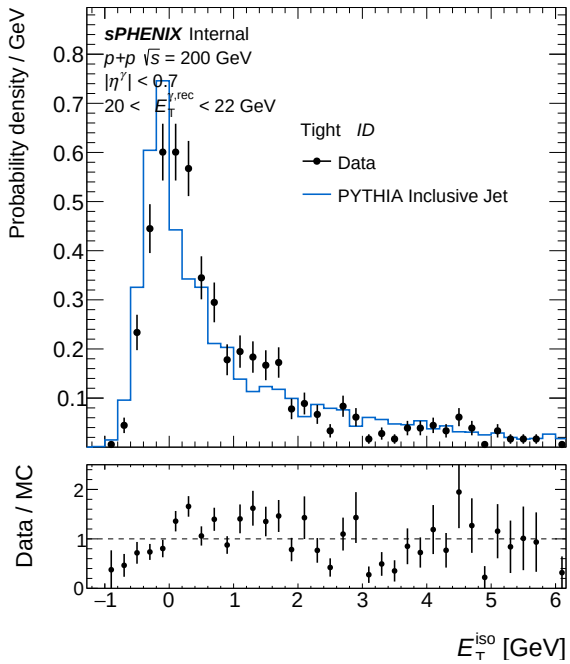

(b) IsoET Tight, pT bin 4 — bdt_npb03



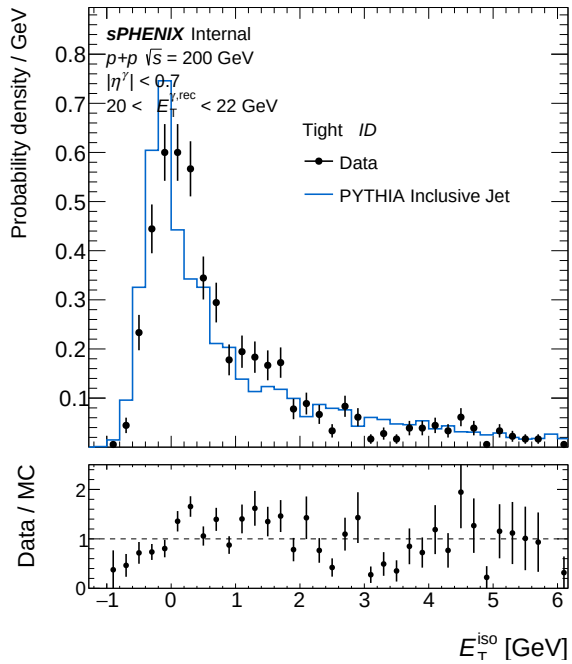

(a) IsoET Tight, pT bin 5 — bdt_nom



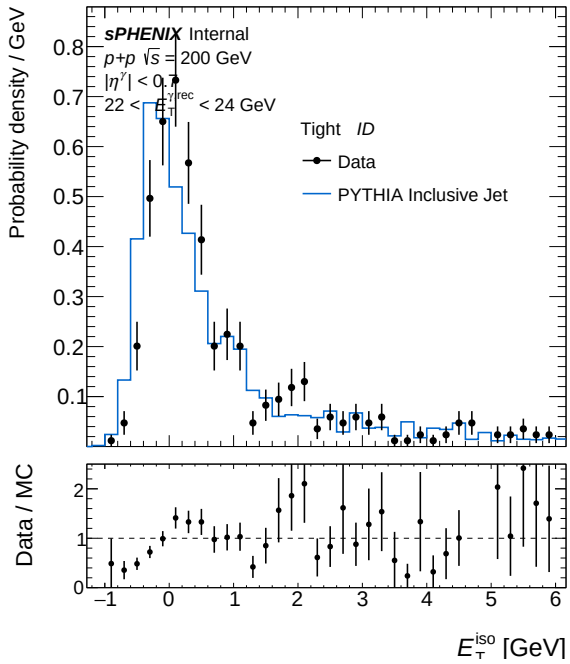

(b) IsoET Tight, pT bin 5 — bdt_npb03



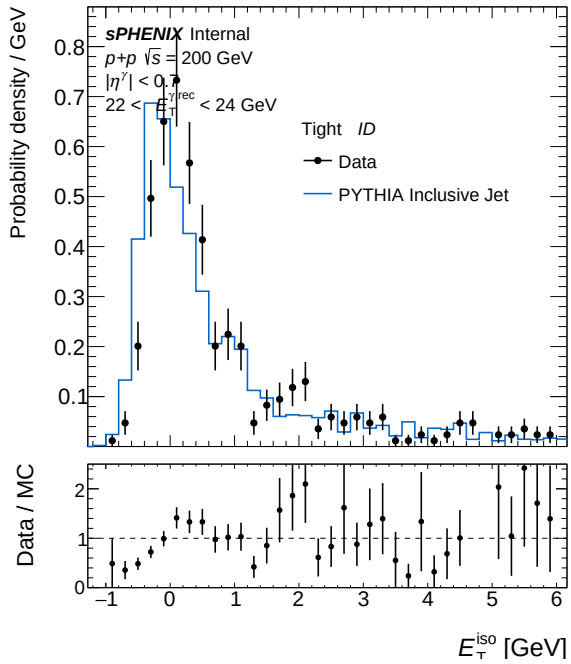

(a) IsoET Tight, pT bin 6 — bdt_nom



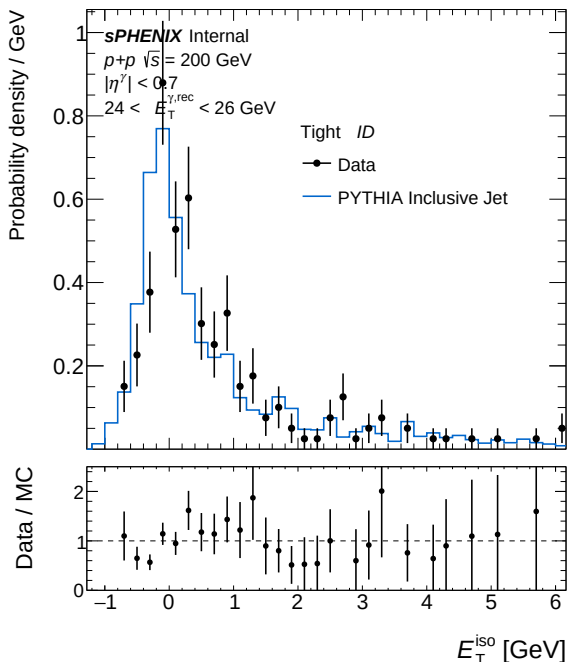

(b) IsoET Tight, pT bin 6 — bdt_npb03



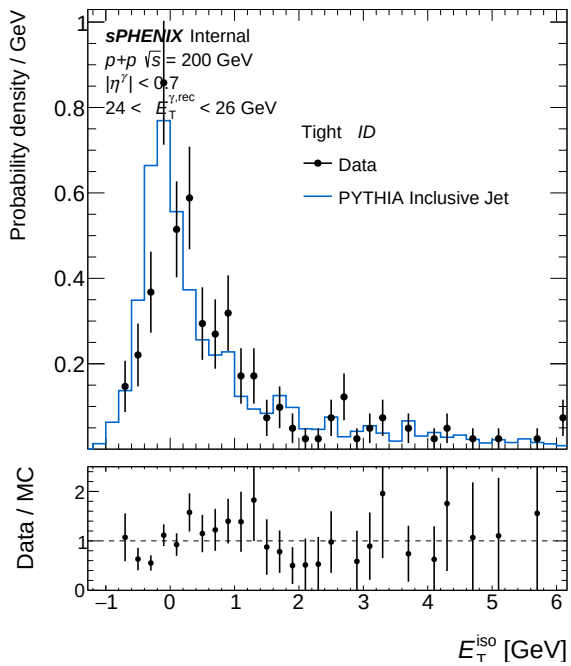

(a) IsoET Tight, pT bin 7 — bdt_nom



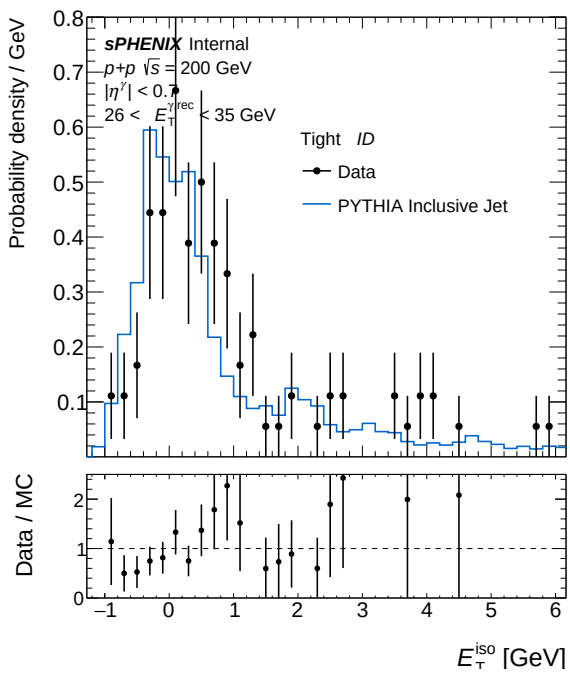

(b) IsoET Tight, pT bin 7 — bdt_npb03



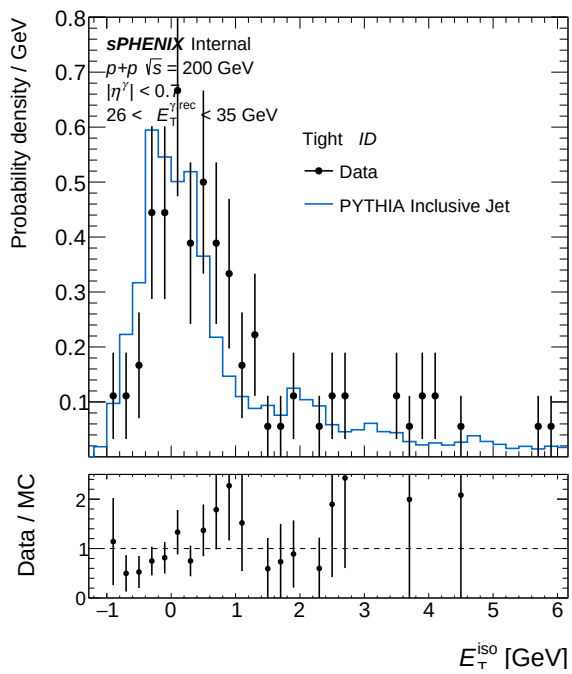

(a) IsoET Tight, pT bin 8 — bdt_nom




(b) IsoET Tight, pT bin 8 — bdt_npb03




(a) IsoET Tight, pT bin 9 — bdt_nom




(b) IsoET Tight, pT bin 9 — bdt_npb03